

Final Report for JPMorgan Chase MLCOE internship – Question 1

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Introduction

Literature Review

The Limitations of "Plug" Variables in Financial Forecasting

Financial statement forecasting is a foundational practice in corporate finance, yet traditional methodologies often compromise the integrity of the double-entry accounting system. As reviewed in Pareja et al. (2007) and Pareja (2009), a significant portion of existing literature relies on the use of "plugs": balancing variables forced to reconcile the difference between Assets and Liabilities plus Equity. Typically, analysts assign a specific account (often Cash, Debt or Equity [\[Pareja\(09\)\]](#)) to absorb any discrepancies in the forecast.

While this approach ensures a balanced sheet mathematically, it is methodologically flawed for robust forecasting. By using a plug, the analyst effectively overrides the natural evolution of the balance sheet, obscuring the causal relationships between operating decisions and financial health. More critically, the use of a plug erodes the primary validation mechanism of the double-entry principle. If the balance sheet is forced to balance, it becomes impossible to detect logical errors or inconsistencies in the underlying code. As noted in the critiques within the Pareja framework, a model that balances by force rather than by logic fails to serve as a reliable tool for stress testing or error detection.

Financial statements are governed by strictly defined accounting identities and deterministic valuation rules, such as the First-In, First-Out (FIFO) inventory method [\[IASB\(21\)\]](#). Consequently, the direct application of machine learning algorithms, such as Long Short-Term Memory (LSTM) networks, to forecast individual line items is often methodologically flawed. Because these 'black box' models treat variables as probabilistic time series rather than components of an integrated system, they fail to enforce the structural interconnectiveness required by double-entry bookkeeping. This limitation frequently results in forecasts that violate the

fundamental accounting equation ($\text{Assets} = \text{Liabilities} + \text{Equity}$), thereby rendering the model incapable of accurately assessing a company's overall financial health.

Proposed Methodology: A Structural "No-Plug" Approach

To address these limitations, this report implements a structural forecasting model based on the framework proposed by Pareja (2009). The core distinction of this method is the rejection of circular references and balancing plugs. Instead, the model relies on a fully integrated system where the Balance Sheet is derived endogenously from the Cash Budget and Income Statement.

In this approach, every line item is driven by specific causal factors (e.g., Receivables driven by Sales, Payables driven by Operating Expense) and the calculation of debt, and the ending cash balance is a calculated result of these flows rather than a residual plug.

In Pareja (2009)'s framework, all the policy variables are known, and the models are used to forecast the balance sheet based on predicted sales and purchases accounting for pricing and volume relationship and inflation. In our case, we do not know the policy variables, but know a company's past financial statements. Using simple linear equations and machine learning algorithms, we train the policy variables that will satisfy the historical financial data. We initially assume that the variables remain constant within our training data, and in later sections apply machine learning methods to account for probability distribution of certain variables to provide richer predictions.

We have further refined the Pareja (2009) framework to better reflect the treasury strategies of modern multinational corporations like Apple or General Motors. Where the original model might treat Marketable Securities solely as a reservoir for excess cash, our implementation models these securities as a distinct portion of the company's liquidity strategy [Almeida(14)]. This adjustment allows for a more realistic depiction of how large firms manage capital allocation between operational cash and interest-bearing assets.

Moreover, Pareja (2009) assumes long-term debt to always have 10-year maturity. In reality, various debt products with varying maturity schedules exist. We, therefore, use an average maturity value as a variable to be also trained.

Once the relationships and policy variables connecting the financial statement elements are established, the methodology utilizes Sales Revenue and Purchases as the central drivers. By anchoring the forecast to these core operational activities, the model can accurately predict the evolution of the balance sheet and income statement while maintaining the necessary accounting linkages.

Testing and Validation Plan

The validity of this implementation is verified through a rigorous "Balance Check." Because the model eschews plugs, the fundamental accounting identity ($\text{Assets} = \text{Liabilities} + \text{Equity}$) serves as a strict test of correctness rather than a guaranteed outcome.

Our testing plan involves running the forecast over a multi-year horizon and calculating the delta between Total Assets and Total Claims at every time step. In this "no-plug" environment, a delta of zero confirms that all cash inflows and outflows have been correctly captured and routed through the Cash Budget. Any deviation from zero indicates a specific error in the logic (e.g., a "leak" in the system where an expense was recorded but cash was not deducted), allowing us to pinpoint and resolve structural flaws that traditional models would mask.

The following sections (**1. The Balance Sheet Elements**, **2. Mathematical Equations Governing the Balance Sheet Elements**, and **3. Time Series Representation of the Balance Sheet**) are answers to the questions 1 and 2, submitted codes are answers to questions 3 and 4, and the remaining sections (**4. Training the Model** and **5. Applying Machine Learning**) provide answers to questions 5, 6, 7, 8. For Part 2, the answers to the questions a) ~ i) are directly mapped to its headings.

Part 1

1. The Balance Sheet Elements

The elements of a balance sheet from Pareja(09) are as follows:

Assets

Net fixed assets (NFA)

Advance payments to suppliers for purchases (AdvPP)

Account receivables (AR)

Inventory (Inventory)

Short-term investments (Invest)

Cash equivalent (Cash)

Liabilities

Account payables (AP)

Advance payments from customers for sales (AdvPS)

Short-term debt (STDebt)

Long-term debt (LTDebt)

Equity

Equity invested (EI)

Cumulated retained earnings (CRE)

Cumulated buyback of equity (CBB)

Net Income (NI)

On the balance sheets of real companies, there are likely to be more elements. For example, there could be current and non-current ‘other liabilities’ other than the 4 mentioned above, and there could be various other assets including deferred tax, non-current account receivables, Goodwill, and other intangible assets. For the purpose of this simple modeling exercise, we first identify the elements listed in the above balance sheet, and lump all unaccounted-for elements as shown below, using GM’s balance sheet as example:

- Assets
 - All unaccounted-for non-interest-bearing elements are lumped with “non-current assets” along with net PPE.
 - * Deferred tax, non-current account receivables, goodwill, other assets
 - * Depreciation rate based on net fixed asset will be reduced accordingly.
 - We use other current assets as the proxy for advance payments to suppliers for purchases.
 - Short-term and long-term investments in market securities are lumped into “investments in market securities.”
 - * Earn a combined average interest.
 - We also combine cash equivalents with investments in market securities to for “total liquidity” because companies do not just hold onto a pile of cash, but rather invest a significant portion of cash in securities to earn interest, which is different from [Pareja\(09\)](#)’s logic of only investing in excess cash after meeting a minimum cash balance.
 - * Total liquidity = cash equivalent + investments in market securities
- Liabilities
 - Advance payments from customers to be fulfilled next year are represented as deferred revenue in company balance sheets.

- All unaccounted-for elements are lumped with long-term debt, to be labeled “non-current liabilities”
 - * Includes non-current long-term debt, employee benefits, non-current deferred revenue, non-current accrued expenses, long-term capital lease obligation
 - These other non-current liabilities are like interest-free long-term debt
 - The total long-term liability is assumed to have the average maturity of AvgM years, and it generates an average LT interest (AvgLTInt) owed by the company
 - * Something important to note is that the “long-term debt” in an actual balance sheet EXCLUDES the principal owed next year.
 - $\text{Non-current liabilities}_t = \text{Total LT liability}_t * (1 - 1/\text{AvgM})$
- = $\text{Total LT liability}_t * (\text{AvgM} - 1) / \text{AvgM}$
- We group all other liabilities due next year as “current liabilities.”
 - Includes STDebt, current accrued expenses which include dealer and customer allowances, claims and discounts, product warranty, payrolls and benefits, and others, to be considered as “short-term loan”
 - * Accrued expenses are like interest-free short-term debt.
 - * Short-term loan (STLoan_t) generates an average ST interest (AvgSTInt_t) owed by the company
 - Also includes current debt which includes principal payment from short-term debt and ALSO INCLUDES the current portion of the total LT liability
 - * $\text{Current liabilities}_t = \text{Total ST liability}_t + \text{Total LT liability}_t / \text{AvgM}$
- = $\text{Total ST liability}_t + \text{Non-current liabilities}_t / (\text{AvgM} - 1)$
- Equity
 - Represented by Stockholders Equity (SE_t)
 - * $\text{SE}_t = \text{SE}_{t-1} + \text{EF}_t + \text{NI}_t - \text{Dividends}_{t-1} - \text{BB}_t$
 - SE_{t-1} = stockholder equity last year
 - EF_t = equity financing to cover portion of CapEx spending
 - NI_t = net income this year
 - Dividends_{t-1} = Dividends earned last year, paid this year
 - BB_t = stock buyback this year

So, our updated balance sheet, along with subitems for clarity, becomes:

Assets

Non-current assets (**NCA_t**)

- Net PPE, deferred tax, non-current account receivables, goodwill, other assets

Advance payments to suppliers for purchases (**AdvPP_t**)

Account receivables (**AR_t**)

Inventory (**Inv_t**)

Total liquidity (**TL_t**)

- Investments in market securities (**IMS_t**)

– Short-term and long-term investments

- Cash equivalent (**Cash_t**)

Liabilities

Account payables (**AP_t**)

Advance payments from customers for sales (**AdvPS_t**)

Non-current liabilities (**NLiab_t**)

- Non-current employee benefits, non-current deferred revenue, non-current accrued expenses, long-term debt and capital lease obligations MINUS next year's principal, other non-current liabilities
- **NLiab_t** = Total LT liability_t * (1 - 1/**AvgM**)

Current liabilities (**CLiab_t**)

- ST debt, current accrued expenses, other current liabilities, and current portion of the long-term debt and capital lease obligations principals
- **STLoan_t** = ST debt_t + current accrued expenses_t + other current liabilities_t
- **CLiab_t** = **STLoan_t** + Total LT liability_t / **AvgM** = **STLoan_t** + **NLiab_t** / (**AvgM** - 1)

Equity

Stockholder equity (**SE_t**)

- **SE_t** = **SE_{t-1}** + **EF_t** + **NI_t** - **Dividends_{t-1}** - **BB_t**
 - SE_{t-1} : total stockholder equity last year
 - EF_t : equity financing to cover portion of CapEx spending
 - NI_t : net income this year
 - Dividends_{t-1} = Dividends earned last year, paid this year
 - BB_t : stock buyback this year

Post Part 1:

We further upgrade the models to better represent Apple's historical data, namely the effective ST debt in Current liabilities:

Assets

Same as before

Liabilities

Account payables (**AP_t**)

Advance payments from customers for sales (**AdvPS_t**)

Non-current liabilities (**NLiab_t**)

- Non-current employee benefits, non-current deferred revenue, non-current accrued expenses, long-term debt and capital lease obligations MINUS next year's principal, other non-current liabilities
- **NLiab_t** = Total LT liability_t * (1 - 1/**AvgM**)

Current liabilities (**CLiab_t**)

- Consists of the operational short-term obligations (Effective ST Debt) plus the principal due this year on long-term loans.

- Effective ST Debt (**EffSTDebt_t**): ST debt (commercial paper), current accrued expenses, and other current liabilities.
 - Modeled as a logit-linear policy ratio of sales: $\text{EffSTDebt}_t = \text{Sales}_t * \%STDebt_t$
- Current LT Debt (**CurrLTDebt_t**): Current portion of the long-term debt and capital lease obligations principals.
 - $\text{CurrLTDebt}_t = \text{Total LT liability}_t / \text{AvgM} = \text{NLiab}_t / (\text{AvgM} - 1)$
- Total Current Liabilities Calculation: $\text{CLiab}_t = \text{EffSTDebt}_t + \text{CurrLTDebt}_t$

Equity

Same as before

2. Mathematical Equations Governing the Balance Sheet Elements

We refer to [Pareja\(09\)](#) for constructing the mathematical equations that link all the elements on the balance sheet without circularity or plugs. First, we define a single primary driver that will predict the evolution of the balance sheet’s elements.

- Sales_t : the total revenue generated by the company in year t

This is a different approach from [Pareja\(09\)](#) where they modeled the unit price, the unit cost, and the volume sold based on market research data on volume’s price sensitivity. For the scope of this problem set, we lump these variables into an overarching sales driver whose evolution is modeled independently based on historical data.

We begin by modeling the asset elements.

1. Assets

1. *Non-Current Assets (NCA)*

The balance of non-current assets evolves sequentially as existing fixed assets depreciate over their useful lifecycle and new capital expenditures (CapEx) are capitalized. The fundamental roll-forward equation is defined as:

$$\text{NCA}_t = \text{NCA}_{t-1} - \text{Depreciation}_t + \text{CapEx}_t \quad (1)$$

Depreciation is modeled as a percentage of the beginning-of-period non-current asset base:

$$\text{Depreciation}_t = \text{NCA}_{t-1} \cdot \%Depreciation \quad (2)$$

Note: Because our model aggregates all Non-Current Assets, which may include non-depreciable assets such as land or goodwill, the trained effective depreciation rate (%Depr) will naturally be lower than the strict accounting depreciation rate applied purely to Plant, Property, and Equipment (PP&E).

To model CapEx, we diverge from a rigid steady-state assumption where maintenance perfectly offsets depreciation. Instead, we decompose CapEx into a maintenance component (scaled by a trainable parameter, %AM, to reflect partial, full, or inflationary replacement) and a growth component driven by sales expansion:

$$\text{CapEx}_t = \%AM \cdot \text{Depreciation}_t + \%AG \cdot \text{Sales}_t \quad (3)$$

By substituting (2) and (3) into (1), we can express the net change in non-current assets (ΔNCA_t) purely in terms of our modeled variables. During model training, the optimizer learns the parameters for the effective depreciation rate (%Depr), the maintenance replacement ratio (%AM), and the incremental asset growth coefficient (%AG).

2. *Cost of Goods Sold (COGS) and Inventory*

Instead of modeling purchases independently and using the Cost of Goods Sold (COGS) as a plug, we directly model COGS based on historical margin trends. We define the Cost Ratio (CR) as the proportion of sales consumed by COGS:

$$CR_t = COGS_t / Sales_t \quad (4)$$

To ensure our predicted cost ratio remains strictly bounded between 0 and 1, we transform the historical cost ratio using the logit function and model it as a linear function of time (t):

$$\text{logit}(CR_t) = \ln \left(\frac{CR_t}{1 - CR_t} \right) = \alpha_{CR} + \beta_{CR} \cdot t \quad (5)$$

The parameters α_{CR} and β_{CR} are trained on historical data by minimizing the mean squared error between the historical $\text{logit}(CR)$ and the predicted $\text{logit}(CR)$. Once the future Cost Ratio is predicted, COGS is calculated directly:

$$COGS_t = CR_t \cdot Sales_t \quad (6)$$

Next, inventory can be modeled as a function of last year's inventory, this year's purchases, and this year's COGS:

$$Inv_t = Inv_{t-1} + Purchases_t - COGS_t \quad (7)$$

As done in [Pareja\(09\)](#), we set the inventory target as a percentage of sales:

$$Inv_t = \%Inv \cdot Sales_t \quad (8)$$

By substituting our known variables into (7), we can solve for $Purchases_t$, which now acts as the necessary plug to balance the inventory account:

$$Purchases_t = Inv_t - Inv_{t-1} + COGS_t \quad (9)$$

$$Purchases_t = \%Inv \cdot (Sales_t - Sales_{t-1}) + COGS_t \quad (10)$$

We add %Inventory (%Inv), α_{CR} and β_{CR} to the variables to be trained.

3. Advance Payments for Purchases

Advance payments to suppliers for purchases (AdvPP) can be modeled as strictly proportional to the total purchases made during the current year. This approach relies on the simplifying assumption that current purchasing volumes serve as a reliable proxy for near-term future demand. Therefore, we define:

$$AdvPP_t = \%AdvPP \cdot Purchases_t \quad (11)$$

We add %Advance payments to suppliers (%AdvPP) to the variables to learn in our training.

4. Account Receivables

Next, we have account receivables. A fraction of this year's sales to customers are on credit and is receivable next year:

$$AR_t = \%AR \cdot Sales_t \quad (12)$$

We add %Account receivables (%AR) to the variables to learn in our training.

5. *Total Liquidity*

Finally, the last element in our assets is total liquidity (TL). Following [Pareja\(09\)](#), we model the total liquidity available as a percentage of sales. Within the liquidity, we assume that a portion of it is invested in market securities (IMS):

$$TL_t = \%TL \cdot Sales_t \quad (13)$$

$$Cash_t = \%Cash \cdot TL_t \quad (14)$$

$$IMS_t = (1 - \%Cash) \cdot TL_t \quad (15)$$

We add %TL and %Cash to our variables to be trained.

Before we move onto liabilities, it is helpful to generate a Liquidity Budget to model the flow of cash in and out of cash balance and short-term investment into market securities. We have updated the name from Cash Budget to Liquidity Budget to reflect our use of total liquidity to represent both cash and market securities investment.

2. Liquidity Budget

We can break down flows into/out of liquidity balance into net liquidity balances (NLB) pertaining to the following activities: operating, capital expense, financing, external investment, and transaction with owners. The final liquidity is the sum of all NLBs.

1. *Operating NLB*

Operating activity captures all inflows from sales (current year sales revenue, account receivables from last year, advance payments from customers for next year) and all outflows (current year purchases, account payables from last year, advance payments to suppliers for next year, operating expenses, income tax):

$$Inflows_t = AR_{t-1} + AdvPS_t + Sales_t \cdot (1 - \%AR - \%AdvPS) \quad (16)$$

$$Outflows_t = AP_{t-1} + AdvPP_t + Purchases_t \cdot (1 - \%AP - \%AdvPP) + OpEx_t + IT_t \quad (17)$$

$$Operating\ NLB_t = Inflows_t - Outflows_t \quad (18)$$

where $OpEx_t$ is the operating expense, and IT_t is income tax. We can model $OpEx_t$ as a function of sales and inflation:

$$OpEx_t = \%OR \cdot Sales_t + OB_{t-1} \cdot (1 + Inflation_t) \quad (19)$$

where OB is the base operating expense that increases with inflation each year (rent, insurance), and $\%OR$ is the rate of increase proportional to this year's sales revenue that impacts the variable portion of the operating expense. Inflation data can be obtained from U.S. Bureau of Labor Statistics.

We can model IT_t as a percentage of net income (NI_t):

$$IT_t = \%IT \cdot NI_t / (1 - \%IT) = NI_t / (1/\%IT - 1) \quad (20)$$

We add $\%IT$, OB_{T_start} (base operating expense of the first element in a time series) and $\%OR$ to our variables to be trained.

2. Capital Expense NLB

Investment is made into the non-current assets to:

- a) keep non-current assets price the same by accounting for depreciation
- b) grow non-current assets by some percentage.

We have already done the calculation in (3), which we reproduce here:

$CapEx_t = Depreciation_t + Sales_t \cdot \%Asset \text{ growth from (3)}$

$$CapEx \text{ NLB}_t = -CapEx_t = -Depreciation_t - Sales_t \cdot \%Asset \text{ growth} \quad (21)$$

1. Financing NLB

This is the net liquidity balance after obtaining more liquidity from new short-term loan ($STLoan_t$) and new long-term loan ($LTLoan_t$), and paying all debt obligations this year.

$$Financing \text{ NLB}_t = STLoan_t + LTLoan_t - CLiab_{t-1} - Avg. \text{ ST interest}_t - Avg. \text{ LT interest}_t \quad (22)$$

where current liabilities_{t-1} ($CLiab_{t-1}$) are the principals from short-term liabilities and long-term liabilities paid this year, represented by:

$$CLiab_{t-1} = STLoan_{t-1} + NLiab_{t-1} / (AvgM - 1) \quad (23)$$

The average short-term interest ($AvgSTInt_t$) and average long-term interest ($AvgLTInt_t$) owed are calculated as follows:

$$AvgSTInt_t = \%AvgSTInt \cdot STLoan_{t-1} = \%AvgSTInt \cdot (CLiab_{t-1} - NLiab_{t-1} / (AvgM - 1)) \quad (24)$$

$$AvgLTInt_t = \%AvgLTInt \cdot Total \text{ LT liability}_{t-1} = \%AvgLTInt \cdot (NLiab_{t-1} / (1 - 1/AvgM)) \quad (25)$$

We add $AvgM$, $\%AvgSTInt$ and $\%AvgLTInt$ to our variables to learn. We derive the expression for $STLoan_t$ and $LTLoan_t$ in 2.3 Liabilities.

2. External Investment NLB

We redeem external investments from last year with a fixed percentage of return ($\%MSReturn$), and make new investments this year:

$$External \text{ investment NLB}_t = IMS_{t-1} \cdot (1 + \%MSReturn) - IMS_t \quad (26)$$

Instead of only allocating the excess cash in market securities, we invest a proportion of total liquidity balance. Using (15):

$$External \text{ investment NLB}_t = (1 - \%Cash) \cdot (TL_{t-1} \cdot (1 + \%MSReturn) - TL_t) \quad (27)$$

We add $\%MSReturn$ to our training variables to be learned.

3. Transaction with owners NLB

In [Pareja\(09\)](#)'s model, owners can make additional investments to finance a portion of long-term debt, which counts as an inflow to liquidity. The company pays dividends to owners and can also engage in stock buybacks:

$$\text{Owners transaction NLB}_t = \text{Equity financing}_t - \text{Dividends}_{t-1} - \text{Stock buybacks}_t \quad (28)$$

where

$$\text{Equity financing}_t = \text{LTLoan}_t \cdot \%EF / (1 - \%EF) \quad (29)$$

$$\text{Target Dividends}_{t-1} = \%Payout \text{ ratio} \cdot \text{Net income}_{t-1} \quad (30)$$

Rather than calculating dividends strictly as a fixed percentage of current net income, we apply the Lintner Model to account for dividend smoothing. Corporate management typically prefers stable dividend payouts to avoid signaling financial distress to investors. Therefore, the modeled dividend is represented as a weighted average between a target payout (driven by a baseline `%Payout_Ratio`) and the actual dividends paid in the prior period, regulated by an optimized `Adjustment_Speed` parameter. So, the actual dividends equation becomes:

$$\text{Dividends}_{t-1} = \text{Adj. Speed} \cdot \text{Target Dividends}_{t-1} + (1 - \text{Adj. Speed}) \cdot \text{Dividends}_{t-2} \quad (31)$$

$$\text{Stock buybacks}_t = \%Buyback \text{ ratio} \cdot \text{Depreciation}_t + \text{Any excess cash after meeting liquidity target} \quad (32)$$

For companies like Apple, their revenue generates tremendous amount of cash that they sometimes use the excess cash to buy back shares to spread the earnings to shareholders instead of paying a higher dividend which has tax implications. We add `%Equity financing (%EF)`, `%Payout ratio (%PR)` and `%Buyback ratio (%BB)` to our variables to learn. We will define the equations for `Net incomet` in [2.4. Income Statement](#).

4. Final NLB

We obtain the final net liquidity balance_t by summing all the different activities' NLB_t:

$$\begin{aligned} \text{Final NLB}_t = & \text{Operating NLB}_t + \text{CapEx NLB}_t + \text{Financing NLB}_t \\ & + \text{External investment NLB}_t + \text{Owners transaction NLB}_t \end{aligned} \quad (33)$$

This is how we obtain the net liquidity balance at time t without plugs or circularity. The unknown variables in the equation can be trained because we have the target liquidity balance given by [\(13\)](#):

$$\text{TL}_t = \%TL \cdot \text{Sales}_t \text{ from (13)}$$

1. Liabilities

1. Account Payables

Account payables are modeled as a percentage of total purchases made this year, payable next year to the suppliers:

$$\text{AP}_t = \%AP \cdot \text{Purchases}_t \quad (34)$$

We add `%AP` to our variables to be trained.

2. Advance Payments for Future Sales

Advance payments received from customers for future sales (AdvPS_t) are modeled as a direct proportion of the current year's total sales. This assumes that the volume of upfront customer payments scales linearly with overall revenue generation. Therefore, we define:

$$\text{AdvPS}_t = \% \text{AdvPS} \cdot \text{Sales}_t \quad (35)$$

We add $\% \text{AdvPS}$ to our variables to be trained.

3. Current Liabilities

From Part 1:

To model the current liabilities, we first model the effective short-term loan (STLoan_t) issued this year. Short-term loan covers any excess cash needed to cover operational expenses and last year's short-term loan obligation (principal + interest) due this year and bring the total liquidity to this year's liquidity target. We also account for the return from investment in market securities here because the size of investment is directly proportional to cash, **instead of** investing excess cash after ST and LT loan calculations which is done in [Pareja\(09\)](#):

$$\begin{aligned} \text{Short-term cash deficit}_t = & \text{TL}_t - \text{TL}_{t-1} - \text{Operating NLB}_t \\ & + \text{STLoan}_{t-1} \cdot (1 + \% \text{AvgSTInt}) - \% \text{MSReturn} \cdot \text{IMS}_{t-1} \end{aligned} \quad (36)$$

$$\text{STLoan}_t = \max(0, \text{Short-term cash deficit}_t) \quad (37)$$

Then, we calculate the effective long-term loan (LTLoan_t) issued this year to finance CapEx_t , last year's current portion of the long-term liability, interest on long-term liability, and dividends to bring the liquidity to this year's target after accounting for the equity financing:

$$\begin{aligned} \text{LTLoan}_t = & \max(0, \text{Short-term cash deficit}_t - \text{STLoan}_t - \text{CapEx NLB}_t \\ & + \text{NLiab}_{t-1} / (\text{AvgM} - 1) + \text{AvgLTInt}_t + \text{Dividends}_{t-1} + \text{Stock buybacks}_t) \cdot (1 - \% \text{EF}) \end{aligned} \quad (38)$$

Current liabilities are summation of the effective short-term loan and the current portion of non-current liabilities, and are represented by:

$$\text{CLiab}_t = \text{STLoan}_t + \text{Total LT liability}_t / \text{AvgM} \quad (39)$$

where

$$\text{Total LT liability}_t = \text{LTLoan}_t + \text{NLiab}_{t-1} \quad (40)$$

Post Part 1:

We adjust how we model the current liabilities to better fit Apple's historical data.

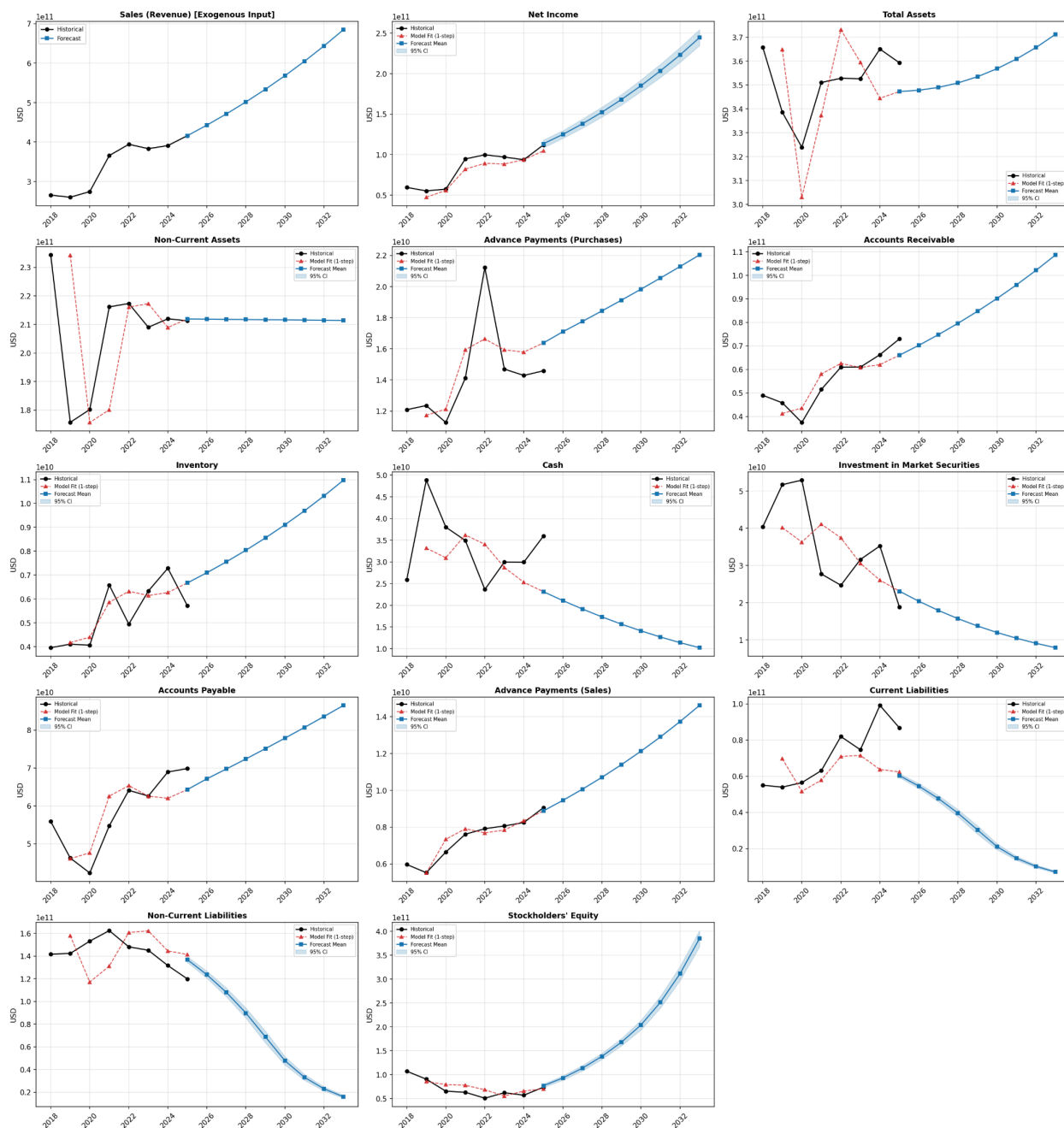
Modeling Effective Short-Term Debt

The Problem

A strictly deficit-driven approach calculates new short-term loans as a reaction to cash shortfalls. This forces short-term borrowing to zero when a company has surplus liquidity. However, in reality, large corporations (e.g., Apple) strategically carry significant commercial paper and revolving credit lines (\$45B–\$88B) despite holding massive cash reserves.

The forecasted Current and non-current liabilities thus fall to zero:

Financial Model: Historical Fit & Monte Carlo Forecast



Defining the Metric

Effective short-term debt isolates the operational and short-term financing liabilities that scale directly with the size of the business, stripping out strict trade payables and the current portion of long-term debt.

The balance sheet identity is defined as:

$$\text{Effective ST Debt}_t = \text{Current Liabilities}_t - \text{Accounts Payable}_t - \text{Advance Payments Sales}_t - \text{Current LT Debt}_t \quad (41)$$

Which equates to the core components we are modeling:

$$\text{Effective ST Debt}_t = \text{ST Debt}_t + \text{Accrued Expenses}_t + \text{Other Current Liabilities}_t \quad (42)$$

The Solution & Implementation

To reflect actual corporate financing behavior, effective short-term debt is now modeled as a policy-driven metric rather than a deficit-driven reaction. It is projected as a logit-linear ratio of sales, mirroring the methodology used for the cost ratio and target liquidity.

1. **Logit-Linear Trend:** We calculate the target short-term debt as a percentage of sales using a sigmoid function over the time index t . This utilizes the learned parameters $\alpha_{\text{ST_Debt}}$ and $\beta_{\text{ST_Debt}}$:

$$\% \text{STDebt}_t = \text{sigmoid}(\alpha_{\text{ST_Debt}} + t \cdot \beta_{\text{ST_Debt}}) \quad (43)$$

2. **Effective ST Debt Calculation:** The actual target volume is then derived directly from the forecasted sales for that period:

$$\text{Effective ST Debt}_t = \text{Sales}_t \cdot \% \text{STDebt}_t \quad (44)$$

3. **Liquidity Cascade:** Because short-term debt is now an established structural policy, it acts as an *input* to the long-term liquidity deficit. The calculated effective short-term debt funds a portion of the required capital, shifting the role of the purely deficit-driven "plug" entirely to the long-term financing variable.

1. *Non-current Liabilities*

Non-current liabilities include the new effective LTLoan_t issued this year, and last year's non-current liabilities minus the current portion of this year's non-current liabilities.

$$\text{NLiab}_t = \text{Total LT liability}_t \cdot (1 - 1/\text{AvgM}) \quad (45)$$

1. Equity

This year's equity accounts for the additional equity financing this year, net income from this year, dividends paid out this year, and stock buyback this year:

$$\text{SE}_t = \text{SE}_{t-1} + \text{EF}_t + \text{NI}_t - \text{Dividends}_{t-1} - \text{BB}_t \quad (46)$$

where

$$\text{EF}_t = \text{LTLoan}_t / (1 - \% \text{EF}) \cdot \% \text{EF} \quad (47)$$

$$\text{Dividends}_{t-1} = \% \text{PR} \cdot \text{NI}_{t-1} \quad (48)$$

$$BB_t = \%BB \cdot \%Depreciation \cdot NCA_{t-1} + \text{any excess cash after meeting liquidity target} \quad (49)$$

We have most of what we need to learn the training variables except dividends and income tax, which will come from modeling the net income in the income statement.

2. Income Statement

We have all the components needed from the previous sections to construct the elements related to net income.

EBITDA_t (earnings before interest, taxes, depreciation, and amortization) can be calculated by:

$$EBITDA_t = Sales_t - COGS_t - OpEx_t \quad (50)$$

Earnings before tax is calculated by subtracting interest payments, depreciation (and amortization):

$$EBT_t = EBITDA_t - Depreciation_t - AvgSTInt_t - AvgLTInt_t + \%MSReturn \cdot IMS_{t-1} \quad (51)$$

Net income (NI_t) is determined by subtracting the income tax:

$$NI_t = EBT \cdot (1 - \%IT) \quad (52)$$

Next year's dividends were already modeled in (30)

Dividends_t = %PR · NI_t from (30)

We now have everything we need to represent every element in the balance sheet as a time series.

3. **Time Series Representation of the Balance Sheet**

We re-write every element in the balance sheet as a time series that have been simplified by combining the equations above. The primary drivers are Sales_t and Purchases_t, which drive every element given the variables to be trained are learned.

1. Variables to be trained

- %AG : Asset growth rate as a percentage of non-current assets
- %AM : Asset maintenance ratio as a percentage of depreciation
- %Depr : Depreciation rate as a percentage of non-current assets
- %AdvPS : Advance payments from customers as a percentage of next year's sales
- %AdvPP : Advance payments to suppliers as a percentage of next year's purchases
- %AR : Account receivables as a percentage of this year's sales
- %AP : Account payables as a percentage of this year's purchases
- %Inv : Inventory target as a percentage of this year's sales
- α_{CR} and β_{CR} : Variables that make up the Cost Ratio via logit model

$$- \text{logit}(CR_t) = \ln (CR_t / (1 - CR_t)) = \alpha_{CR} + \beta_{CR} * t$$
- %TL : Total liquidity as a percentage of this year's sales
- %Cash : Cash as a percentage of total liquidity
- %IT : Income tax as a percentage of net income this year
- %OR : Variable portion of OpEx that increases in proportion to this year's sales

- OB_{T_start} : Baseline OpEx that increases with this year's average inflation
- α_{ST_Debt} and β_{ST_Debt} : parameters for sigmoid function for calculating Effective ST debt
- %AvgSTInt : Average short-term interest on the effective short-term loan
- %AvgLTInt : Average interest on total long-term liabilities
- AvgM : Average maturity of total long-term liability
- %MSReturn : Percentage return on the investment in market securities
- %EF : Equity financing as a percentage of total excess cash needed after accounting for long-term investments and long-term obligations
- %PR : Dividends payout ratio as a percentage of net income this year, paid next year
- Adj. speed : Speed at which to adjust the dividends amount paid next year
- %BB : Stock buyback as a percentage of asset depreciation this year

1. Parameter Constraints and Bounds

To ensure the model generates economically logical and mathematically stable financial statements, strict constraints are enforced on both the trainable variables and the underlying equations. These constraints are implemented natively within the optimizer using TensorFlow Probability Bijections.

1. Domain Constraints on Trainable Variables

- **Strict Positivity ($x > 0$):** Variables that cannot logically be negative, such as the Depreciation Rate (%Depr), Asset Growth (%AG), and Interest Rates (%AvgSTInt, %AvgLTInt), are constrained using a Softplus activation function. This prevents the optimizer from testing negative rates during gradient descent.
- **Percentage Bounds ($0 < x < 1$):** Variables representing fractions of a whole, such as the Income Tax Percentage (%IT), Dividend Payout Ratio (%PR), and the Lintner Dividend Adjustment Speed (α), are constrained using a Sigmoid function.
- **Maturity Boundary ($AvgM > 1.001$):** Because the calculation for the current portion of long-term debt divides by $(AvgM - 1)$, the average maturity must strictly exceed 1 year to prevent mathematical singularity and infinite liability generation. This is enforced using a chained bijection: a Softplus function shifted by a constant of 1.001.
- **Trended Ratios:** Time-varying policies, such as the Cost Ratio (CR) and Effective Short-Term Debt percentage, are modeled as logit-linear trends. By wrapping the linear trend $(\alpha + \beta t)$ in a sigmoid function, the forecasted ratios are strictly bounded between 0 and 1, even if extrapolated decades into the future.

2. Logical Constraints on Equations

- **Non-Negative Borrowing:** A company cannot issue "negative" debt to reduce cash. In the Liquidity Budget equations, the issuance of new short-term and long-term loans is constrained using a maximum function: $New_Loan_t = \max(0, Liquidity_Deficit_t)$. If there is a cash surplus, borrowing falls to exactly zero, and the excess cash cascades into stock buybacks.
- **Structural Identity Check:** The fundamental accounting equation ($Assets = Liabilities + Equity$) is strictly evaluated at every time step t . Rather than enforcing this via a loss function penalty (which would allow minor deviations), it is enforced structurally by the double-entry routing of the Cash Budget. The simulation calculates a delta to verify that the mismatch is exactly zero.

1. Variables from External Environment

Since we assume a flat short-term and long-term interest for loans and return on market securities, the only external variable is inflation.

- Inflation_t : Average inflation at year t
 - 1. Assets
 - Non-current assets
 - $NCA_t = NCA_{t-1} - \text{Depreciation}_t + \text{CapEx}_t$
 - * Where $\text{CapEx}_t = \text{Depreciation}_t + \text{Sales}_t * \%AG$
 - Advance payments to suppliers for purchases
 - $\text{AdvPP}_t = \%AdvPP * \text{Purchases}_t$
 - Account receivables
 - $AR_t = \%AR * \text{Sales}_t$
 - Inventory
 - $\text{Inv}_t = \%Inv * \text{Sales}_t$
 - Total liquidity (**TL_t**)
 - Target $TL_t = \%TL * \text{Sales}_t$
 - * Investments in market securities
 - $IMS_t = (1 - \%Cash) * TL_t$
 - * Cash equivalent
 - $\text{Cash}_t = \%Cash * TL_t$
 - Should be equivalent to: $\text{Final } TL_t = \text{Operating } NLB_t + \text{CapEx } NLB_t + \text{Financing } NLB_t + \text{External investment } NLB_t + \text{Owners transactions } NLB_t$
 - 1. Liabilities
 - Account payables
 - $AP_t = \%AP * \text{Purchases}_t$
 - Advance payments from customers for sales
 - $\text{AdvPS}_t = \%AdvPS * \text{Sales}_t$
 - Current liabilities
 - $CLiab_t$
- $$= \text{STLoan}_t + \text{NLiab}_t / (\text{AvgM} - 1)$$
- where
- $\text{STLoan}_t = \max(0, \text{STCashDeficit}_t)$
 - $\text{STCashDeficit}_t = TL_t - TL_{t-1} - \text{Operating } NLB_t + \text{STLoan}_{t-1} * (1 + \%AvgSTInt) - \%MSReturn * IMS_{t-1}$
 - $\text{STLoan}_{t-1} = CLiab_{t-1} - \text{NLiab}_{t-1} / (\text{AvgM} - 1)$
 - $\text{Operating } NLB_t = AR_{t-1} + \text{AdvPS}_t + \text{Sales}_t * (1 - \%AR - \%AdvPS) - (AP_{t-1} + \text{AdvPP}_t + \text{Purchases}_t * (1 - \%AP - \%AdvPP) + \text{OpEx}_t + \text{IT}_t)$
 - $AP_{t-1} = \%AP * \text{Purchases}_{t-1}$
 - $\text{AdvPP}_t = \%AdvPP * \text{Purchases}_t$
 - $\text{OpEx}_t = \%OR * \text{Sales}_t + \text{OB}_{t-1} * (1 + \text{Inflation}_t)$

- $IT_t = NI_t / (1/\%IT - 1)$

- $AR_{t-1} = \%AR * Sales_{t-1}$

- $AdvPS_t = \%AdvPS * Sales_t$

- Non-current liabilities

- $NLiab_t$

$$= (LTLoan_t + NLiab_{t-1}) * (AvgM - 1) / AvgM$$

where

- $LTLoan_t = (\max(0, STCashDeficit_t - STLoan_t - CapEx_{NLB_t} + NLiab_t / (AvgM - 1) + AvgLTInt_t + Dividends_{t-1} + Stock\ buybacks_t) * (1 - \%EF)$

- $CapEx_{NLB_t} = NCA_{t-1} * \%Depr + \%AG * Sales_t$

- $Dividends_{t-1} = \%PR * NI_{t-1}$

- NI_{t-1} : net income from last year

$$= (1 - \%IT) * (Sales_{t-1} - COGS_{t-1} - OpEx_{t-1} - Depreciation_{t-1} - AvgSTInt_{t-1} - AvgLTInt_{t-1} + \%MSReturn * (1 - \%Cash) * TL_{t-1})$$

- TL_{t-1} = Total liquidity last year.

- $COGS_{t-1} = \%Inv * (Sales_{t-2} - Sales_{t-1}) + Purchases_{t-1}$

- $OpEx_{t-1} = \%OR * Sales_{t-1} + OB_{t-2} * (1 + Inflation_{t-1})$

- $Depreciation_t = NCA_{t-1} * \%Depr$

- $AvgSTInt_t = \%AvgSTInt * (CLiab_{t-t} - NLiab_{t-t} / (AvgM - 1))$

- $AvgLTInt_t = \%AvgLTInt * (NLiab_{t-1} / (1 - 1/AvgM))$

- $Stock\ buybacks_t = \%BB * Depreciation_t + excess\ cash\ after\ meeting\ liquidity\ target$

1. Equity

- Stockholder equity

- SE_t

$$= SE_{t-1} + LTLoan_t / (1 - \%EF) * \%EF + NI_t - \%PR * NI_{t-1} - \%BB * \%Depr * NCA_{t-1} - excess\ cash\ after\ meeting\ liquidity\ target$$

We see in all elements' equations that they are driven by Sales and Purchases and elements from previous years, indicating no circularity or plugs. The net liquidity balance calculation and the new short-term and long-term borrowing equations ensure that the assets and liabilities + equity are increased/decreased the same. Thus, assets = liabilities + equity will always be satisfied. We can check with test values in our simple TensorFlow model.

This forward model has been implemented in Python in **simple_financial_model.py** using data from Apple's financial statements. The financial statements were first obtained from Yahoo finance as suggested by Question 4 (we wrote the script in **yahoofinance.py**), but we could only access past 4 years of data, which is not sufficient for training in the following section. We managed to obtain older financial data by directly pulling from Apple's 10-K filings. The script that we wrote and used for the data extraction is found in **sec-edgar-download-extractor.py**.

The following is the output of the code when run with: `python .\simple_financial_mode.py` :

Year	Assets	Liabilities	Equity	Check
0	352.75B	302.08B	50.67B	0.00
1	367.38B	318.42B	48.96B	0.00
2	366.89B	318.96B	47.94B	0.00
3	372.97B	325.48B	47.49B	0.00

Even without using any plugs, we see that the assets = liabilities + equity identity is satisfied because our model follows the double-entry principle with any inflows and outflows on the balance sheet as done by Pareja(09).

4. Training the Model

Out of the variables to be trained, there are quite a few that are easily trained independently from other variables by looking at balance sheets and income statements. We first note the primary driver for our model:

- Sales_t : Total revenue in balance sheet

We can train the rest of the variables like the following:

- %AG, %AM : Learn from change in all non-current assets in balance sheet
- %Depr : Learn from depreciation in income statement and non-current assets in balance sheet
- α_{CR} , β_{CR} : Learn from COGS and Sales data
- %AR : Learn from Sales and account receivables data in balance sheet
- %AP : Learn from Purchases and account payables data in balance sheet
- %AdvPS : Learn from current deferred revenue data and current year's Sales in balance sheet
- %AdvPP : Learn from Purchases and current other assets in balance sheet
- %Inv : Learn from inventory data in balance sheet
- %Cash : Learn from short-term investments in balance sheet
- %TL : Learn from revenue and Cash Equivalents And Short Term Investments in balance sheet
- %IT : Learn from net income and tax provision in income statement
- %OR, OB_{T_start} : Learn from inflation data and operating expense in income statement
- %PR, Adj. Speed : Learn from common stock dividends in Cash Flow and net income in income statement
- %BB : Learn from Repurchase Of Capital Stock and depreciation in Cash Flow

Each of the variables above can be related to an element from Cash Flow, Balance Sheet, or Income Statement, and can be trained with a simple mean-square-error minimization for each.

The following variables (i.e. structural parameters) can be trained together using the above trained variables:

- %AvgSTInt : Average short-term interest on the effective short-term loan
- %AvgLTInt : Average interest on total long-term liabilities
- AvgM : Average maturity of total long-term liability
- %MSReturn : Percentage return on the investment in market securities
- %EF : Equity financing as a percentage of total excess cash needed after accounting for long-term investments and long-term obligations

Using net income, interest payments, return from investment in market securities, stockholders equity and current and non-current liabilities data, we can train AvgM, %AvgSTInt, %AvgLTInt, %MSReturn, and %EF concurrently via simple gradient while minimizing mean-square error between predicted and actual net income, stockholders equity, current liabilities, and non-current liabilities.

The code has been implemented in trainable_financial_model.py. For this simple training model, %AM = 100% and Purchases are also used together with Sales as dual driver.

In order to test the model's prediction capability, we train the parameters from Apple's 2018~2024 data, predict 2025 data, and compare it with the actual data. The following is the output after running **trainable_financial_model.py**:

```

Simple Policy Trained Variables
Final %AG: 0.00000
Final %Depr: 0.05553
Final %AdvPS: 0.02006
Final %AdvPP: 0.07579
Final %AR: 0.15866
Final %AP: 0.30793
Final %Inv: 0.01603
Final %TL: 0.19904
Final %Cash: 0.46434
Final %IT: 0.20516
Final %OR: 0.23533
Final Baseline OpEx: -30306718213.65
Final %PR: 0.17871
Final %BB: 7.09752

```

```

-----
Structural Trained Variables
Final %AvgSTInt: 0.13186
Final %AvgLTInt: 0.01496
Final AvgM: 3.26387
Final %MSReturn: 0.00000
Final %EF: 0.22536
-----

```

```

=====
BACKTESTING: Forecasted 2025 vs Actual 2025 Balance Sheet & Net Income
=====

```

Item	Actual (B)	Forecast (B)	Error %
nca	211.28	211.99	0.3%
advance_payments_purchases	14.59	16.36	12.2%
accounts_receivable	72.96	66.03	-9.5%
inventory	5.72	6.67	16.7%
investment_in_market_securities	18.76	44.37	136.5%
cash	35.93	38.46	7.0%
accounts_payable	69.86	63.96	-8.5%
advance_payments_sales	9.05	8.88	-1.9%
current_liabilities	86.72	68.74	-20.7%
non_current_liabilities	119.88	155.62	29.8%
equity	73.73	86.68	17.6%
net_income	112.01	103.07	-8.0%

```

=====
Balance Sheet Check: -0.00

```

```

Liquidity Check: 0.00

```

It shows a modest prediction ability, which is unsurprising because we have lumped various balance sheet elements together to form several elements to be more consistent with models described in [Pareja\(07\)](#) and [Pareja\(09\)](#), as we have explained in **1. The Balance Sheet Elements**. Also, there is a limitation of assuming a liquidity balance (cash + investment in market securities) that varies linearly with sales, especially for a cash-rich company like Apple whose cash balance and stock buybacks are governed by the recent buyback programs in order to reduce its cash balance instead of being proportional to asset

depreciation, which is done here as done by [Pareja\(09\)](#).

Our model can also forecast earnings because net income is automatically calculated when predicting the elements on the balance sheet, and is outputted as “net_income” as shown above.

5. Applying Machine Learning

As our model is rather a simple model based on [Pareja\(09\)](#), there is a lot of improvements that could make the model more performant. A short list of potential improvements is as follows:

- Predicting purchases and sales of future years using Long Short-Term Memory and macroeconomic indicators (e.g., inflation, interest rate)
- Implementing Bayesian model to account for probability distribution of accounting variables (e.g., company policy for capital expenditure, operating expense prediction, cash balance targets) to account for the noise term, $n(t)$.
- Implementing neural networks for each accounting variables that take into consideration macroeconomic signals (e.g., incorporate consumer price index, GDP growth, and interest rates to predict advance purchases percentage)
- Incorporating the company’s stock buyback target if it has been disclosed publicly into training the buyback ratio with a neural network

For our model, we choose to implement Bayesian learning with Variational Inference method [\[Dürr\(20\)\]](#) to train the operational expense because

- a. OpEx is influential in the net income value
- b. OpEx, in our model, is heavily influenced by sales revenue, which is one of the main drivers in our model, up to 23%

Applying Variational Inference allows us to model the aleatoric and epistemic uncertainties of OpEx by calculating the posterior distribution of OpEx’s linear regression parameters given the past OpEx, sales, and inflation data instead of a singular set of parameters (i.e., slope and offset) that a simple linear regression gives you. It allows us to predict the OpEx, and therefore the earnings and other balance sheet elements, with a confidence interval.

Most of the code from `trainable_financial_model.py` remains unchanged except how we model the OpEx during training. The following updates are made:

1. Bayesian Conversion: Replaced the deterministic %OR and OB_{T_start} variables with Variational Posteriors.
 - We learn a Mean (μ) and a Standard Deviation(σ) for each of these variables.
 - During training/inference, we sample their values from their distributions:
 - $Value = \mu + \sigma * \epsilon$, where $\epsilon \sim \mathcal{N}(0, 1)$
2. Loss Function: Replaced the simple Mean Squared Error (MSE) for OpEx with Negative Log Likelihood + KL Divergence.
 - This balances fitting the data (Likelihood) with keeping the distribution close to our prior beliefs (KL).
3. Monte Carlo Simulation: Added a `run_monte_carlo_forecast` function. Instead of predicting one single Net Income line, we run the forecast 1,000 times, sampling different OpEx values each time, to generate Confidence Intervals.

So, to answer the ‘Hint’, we are improving the model such that when we predict the earnings (i.e. net income), we can now model $n(t)$ with the confidence intervals given by the Monte Carlo simulation. After we have trained all the variables, they become the very function that take in this year’s $y(t)$ and input $x(t)$ to predict $y(t+1)$. The input $x(t)$ includes the primary drivers, which are sales and purchases time series,

and inflation, which is the external environment variable that we include in the model for predicting the operating expense.

We implemented the updated Bayesian model in `trainable_financial_model_taxanomaly_effstdebt_bayesian_v1.py` with the `include_tax_anomalies = False`, and the output is reproduced below:

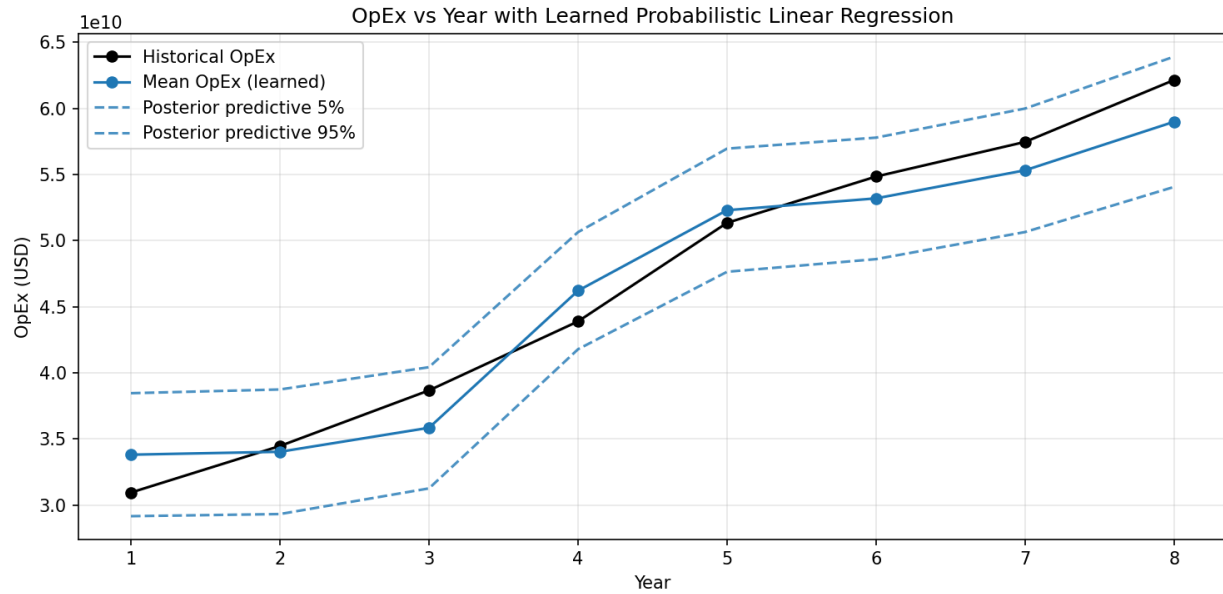
```
Simple parameters:

Final %AG: 0.00000
Final %AM: 0.94062
Final %Depr: 0.05553
Final %AdvPS: 0.02126
Final %AdvPP: 0.07840
Final %AR: 0.15866
Final %AP: 0.30794
Final %Inv: 0.01603
Total Liquidity (baseline + logit-linear): baseline=0.6013, alpha=-2.3463, beta=-0.411579
=> %TL at t=0: 0.0874, %TL at t=7: 0.0053
Cash % of Liquidity (logit-linear): alpha=-0.2235, beta=0.032385
=> %Cash at t=0: 0.4444, %Cash at t=7: 0.5008
Final %IT: 0.17021
Final %PR: 0.99969
Final DivAdjSpeed (alpha): 0.00371
Stock Buyback (baseline + ratio*depr): baseline=1.5540, ratio=-6.610795
Effective ST Debt % of Sales (logit-linear): alpha=-1.7458, beta=0.059738
=> %EffSTDebt at t=0: 0.1486, %EffSTDebt at t=7: 0.2095
Cost Ratio (logit-linear): alpha=0.3459, beta=-0.0475
=> CR at t=0: 0.5856, CR at t=7: 0.5034
Bayesian OpEx Variable %: Mean=0.0923, Std=0.0155
Bayesian OpEx Baseline (USD): Mean=3.91e+10, Std=8.09e+08
OpEx aleatoric uncertainty (USD): 2.49e+09
```

```
Structural parameters:

Final %AvgSTInt: 0.03970
Final %AvgLTInt: 0.00692
Final AvgM: 15.36485
Final %MSReturn: 0.01119
Equity Financing % (logit-linear): alpha=0.5389, beta=0.018432
=> %EF at t=0: 0.6316, %EF at t=7: 0.6610
```

The Operation Expense learned with Bayesian Variational Inference is as follows:

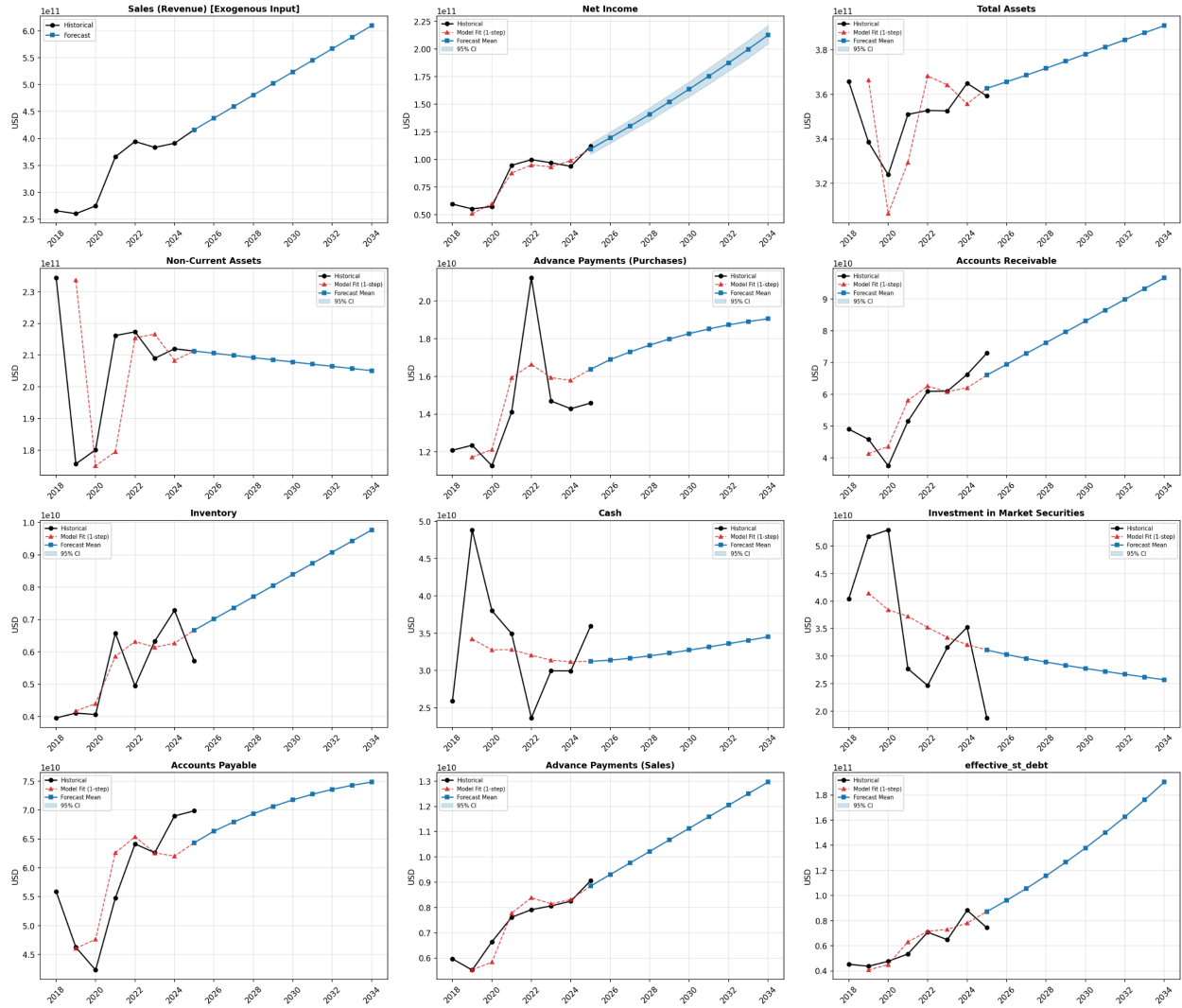


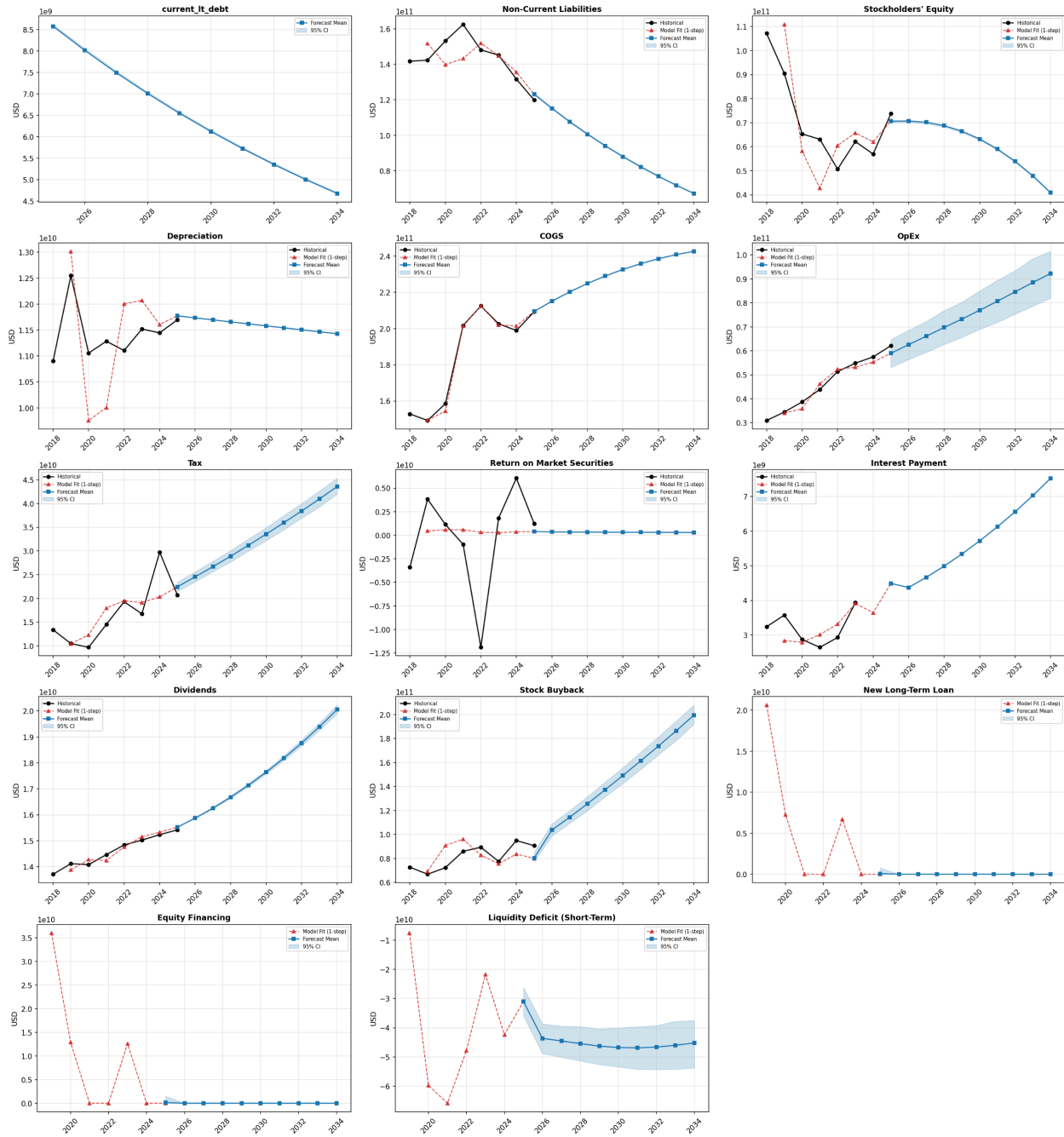
And the net income now has a confidence level due to the noise term, and we see that the uncertainty increases the further the forecast year as expected for Variational Inference model's increasing epistemic uncertainty outside of the training range:

Net Income

Year	Mean	2.5% CI	97.5% CI
2025	1.09e+11	1.04e+11	1.14e+11
2026	1.20e+11	1.15e+11	1.25e+11
2027	1.30e+11	1.25e+11	1.35e+11
2028	1.41e+11	1.35e+11	1.47e+11
2029	1.52e+11	1.46e+11	1.58e+11
2030	1.63e+11	1.57e+11	1.70e+11
2031	1.75e+11	1.68e+11	1.81e+11
2032	1.87e+11	1.80e+11	1.95e+11
2033	2.00e+11	1.92e+11	2.07e+11
2034	2.12e+11	2.04e+11	2.20e+11

Financial Model: Historical Fit & Monte Carlo Forecast





6. Projected Balance Sheet, Income Statement, and Cash Budget

We describe how we calculate the cash budget and income statement which are related to the balance sheet.

We can start with the revenue on the income statement which appears at the top.

Revenue is equivalent to $Sales_t$.

It is followed by operating costs, which include:

- Cost of goods sold (COGS)
- Operational expenses (OpEx such as administrative and sales)
- Depreciation and amortization

EBIT (earnings before interests and taxes) is calculated by

$$\text{EBIT} = \text{Revenue} - \text{COGS} - \text{OpEx} - \text{Depreciation}$$

Then, we account for interest payments and ST investment returns, which are used to calculate EBT (earnings before taxes):

$$\text{EBT} = \text{EBIT} - \text{interest payments} + \text{ST returns}$$

Then, we assess the income tax by multiplying EBT by the effective tax rate:

$$\text{Income Taxes} = \text{EBT} * \% \text{EffTax}$$

Then, net income is found by subtracting the income taxes from EBT:

$$\text{Net Income} = \text{EBT} - \text{Income Taxes}$$

From the net income, we allocate a percentage of it for dividends to be paid out next year, and the rest goes to cumulated retained earnings (CRE) which is part of equity:

$$\text{Next year dividends} = \text{Net Income} * \% \text{Payout}$$

$$\text{CRE}_t = \text{CRE}_{t-1} + \text{Net Income}$$

Cash Budget has 5 modules: Operating Activities, Investing Activities, External Financing, Transactions with Owners, and Discretionary Transactions.

Module 1: Operating Activities

Cash Inflows

Inflows from sales

$$\text{Total inflows} = \text{Inflows from sales}$$

Cash Outflows

Payments for purchases = COGS

Operational Expenses = OpEx

Income Tax

$$\text{Total outflows} = \text{COGS} + \text{OpEx} + \text{Income Tax}$$

Operating Net Cash Balance

$$= \text{Total inflows} - \text{Total outflows}$$

$$= \text{Sales} - \text{COGS} - \text{OpEx} - \text{Income Tax}$$

Module 2: Investing Activities

Cash Inflow

$$\text{Total inflows} = 0$$

Cash Outflow

Investment in fixed assets = CapEx

$$\text{Total outflows} = \text{CapEx}$$

Investing Net Cash Balance

$$= \text{Inflow} - \text{Outflow}$$

$$= -\text{CapEx}$$

Module 3: External Financing

Cash Inflows

ST loan

LT loan

$$Total\ inflows = ST\ loan + LT\ loan$$

Cash Outflows

ST principal payment

ST interest

LT principal payment

LT interest

$$Total\ outflows = ST\ principal + ST\ interest + LT\ principal + LT\ interest$$

Financing Net Cash Balance

$$= Inflow - Outflow$$

$$= ST\ loan + LT\ loan - ST\ principal - LT\ principal - ST\ interest - LT\ interest$$

Module 4: Transactions with Owners

Cash Inflows

Invested equity = Equity financing

$$Total\ Inflows = Invested\ equity$$

Cash Outflows

Dividends from last year

Stock Buyback

$$Total\ Outflows = Dividends\ from\ last\ year + Stock\ buyback$$

Owners' Transaction Net Cash Balance

$$= Inflows - Outflows$$

$$= Equity\ financing - Dividends\ from\ last\ year - Stock\ buyback$$

Module 5: Discretionary Transactions

Cash Inflows

Return from ST investments

$$Total\ inflows = Return\ from\ ST\ investments$$

Cash Outflows

No outflow because our model's policy allocates a fraction of total liquidity to ST investments instead of allocating excess cash to it

Discretionary Transaction Net Cash Balance

$$= Inflows - Outflows$$

$$= Return\ from\ ST\ investments$$

The forecasted 10 years (with inflation = 3%, sales growing linearly at the historical rate of training data) of Balance Sheet, Income Statement, and Cash Budget is attached below:

FORECAST BALANCE SHEET — Mean across Monte Carlo samples (USD)

Line Item	FY2025	FY2026	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032	FY2033	FY2034
ASSETS										
Non-Current Assets	\$211,294,134,686	\$210,597,581,775	\$209,903,333,641	\$209,211,382,686	\$208,521,721,334	\$207,834,342,035	\$207,149,237,266	\$206,466,399,525	\$205,785,821,338	\$205,107,495,254
Advance Payments (Purchases)	\$16,374,224,975	\$16,890,984,352	\$17,292,784,103	\$17,655,445,055	\$17,979,604,046	\$18,265,980,509	\$18,515,372,659	\$18,728,653,189	\$18,906,764,530	\$19,050,713,718
Accounts Receivable	\$66,026,078,355	\$69,438,659,583	\$72,851,240,810	\$76,263,822,038	\$79,676,403,265	\$83,088,984,493	\$86,501,565,720	\$89,914,146,948	\$93,326,728,175	\$96,739,309,403
Inventory	\$6,670,655,799	\$7,015,431,005	\$7,360,206,211	\$7,704,981,417	\$8,049,756,622	\$8,394,531,828	\$8,739,307,034	\$9,084,082,240	\$9,428,857,446	\$9,773,632,652
Cash	\$31,223,669,354	\$31,387,036,118	\$31,642,257,505	\$31,963,551,974	\$32,331,912,553	\$32,733,440,586	\$33,158,028,588	\$33,598,350,511	\$34,049,103,448	\$34,506,446,402
Investment in Market Securities	\$31,123,361,807	\$30,289,228,865	\$29,562,470,016	\$28,911,034,982	\$28,312,312,782	\$27,760,509,142	\$27,214,687,305	\$26,697,337,455	\$26,193,348,602	\$25,699,278,206
TOTAL ASSETS	\$362,712,124,977	\$365,618,921,698	\$368,612,292,286	\$371,710,218,151	\$374,871,710,602	\$378,067,788,593	\$381,278,198,571	\$384,488,969,867	\$387,690,623,539	\$390,876,875,635

LIABILITIES										
Accounts Payable	\$64,316,968,943	\$66,346,768,633	\$67,925,013,842	\$69,349,524,208	\$70,622,801,190	\$71,747,670,678	\$72,727,268,012	\$73,565,021,081	\$74,264,631,697	\$74,830,055,432
Advance Payments (Sales)	\$8,846,925,987	\$9,304,182,488	\$9,761,438,989	\$10,218,695,489	\$10,675,951,990	\$11,133,208,491	\$11,590,464,992	\$12,047,721,492	\$12,504,977,993	\$12,962,234,494
Effective ST Debt	\$87,204,056,424	\$96,117,022,855	\$105,620,475,218	\$115,735,721,110	\$126,483,234,187	\$137,882,470,325	\$149,951,679,605	\$162,707,716,402	\$176,165,850,486	\$190,339,579,048
Current LT Debt	\$8,547,079,717	\$7,992,201,779	\$7,473,346,616	\$6,988,175,623	\$6,534,502,017	\$6,110,280,982	\$5,713,600,453	\$5,342,672,494	\$4,995,825,244	\$4,671,495,379
Non-Current Liabilities	\$123,108,130,835	\$115,115,929,056	\$107,642,582,440	\$100,654,406,817	\$94,119,904,800	\$88,009,623,818	\$82,296,023,365	\$76,953,350,871	\$71,957,525,628	\$67,286,030,248
TOTAL LIABILITIES	\$292,023,161,905	\$294,876,104,811	\$298,422,857,104	\$302,946,523,246	\$308,436,394,184	\$314,883,254,294	\$322,279,036,426	\$330,616,482,341	\$339,888,810,642	\$350,089,394,601

EQUITY										
Equity	\$70,688,963,072	\$70,742,816,887	\$70,189,435,182	\$68,763,694,904	\$66,435,316,418	\$63,184,534,300	\$58,999,162,146	\$53,872,487,527	\$47,801,812,898	\$40,787,481,034

TOTAL LIAB + EQUITY	\$362,712,124,977	\$365,618,921,698	\$368,612,292,286	\$371,710,218,151	\$374,871,710,602	\$378,067,788,593	\$381,278,198,571	\$384,488,969,867	\$387,690,623,539	\$390,876,875,635
CHECK: Assets-(L+E)	\$-0	\$-0	\$-0	\$0	\$-0	\$-0	\$-0	\$0	\$-0	\$-0

FORECAST INCOME STATEMENT — Mean across Monte Carlo samples (USD)

Line Item	FY2025	FY2026	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032	FY2033	FY2034
Revenue (Sales ₁)	\$416,161,000,000	\$437,670,428,571	\$459,179,857,143	\$480,689,285,714	\$502,198,714,286	\$523,708,142,857	\$545,217,571,429	\$566,727,000,000	\$588,236,428,571	\$609,745,857,143
COGS	\$209,479,943,818	\$215,111,448,777	\$220,236,683,361	\$224,862,675,415	\$228,997,548,241	\$232,650,478,706	\$235,831,648,594	\$238,552,189,724	\$240,824,123,455	\$242,660,295,213
OpEx	\$59,189,357,476	\$62,743,853,543	\$66,268,928,527	\$69,970,096,920	\$73,867,187,499	\$77,530,912,397	\$81,309,330,879	\$85,303,190,906	\$89,343,258,569	\$93,310,113,709
Depreciation & Amortization	\$11,772,820,547	\$11,734,009,756	\$11,695,327,383	\$11,656,773,003	\$11,618,346,194	\$11,580,046,536	\$11,541,873,610	\$11,503,826,998	\$11,465,906,283	\$11,428,111,049
EBIT = Revenue - COGS - OpEx - Depreciation	\$135,718,878,159	\$148,081,116,495	\$160,978,917,872	\$174,199,740,376	\$187,715,632,351	\$201,946,705,217	\$216,534,718,346	\$231,367,792,372	\$246,603,140,266	\$262,347,337,172
Interest Payments	\$4,494,432,488	\$4,376,763,861	\$4,672,141,362	\$4,994,830,605	\$5,345,430,056	\$5,724,521,177	\$6,132,660,059	\$6,570,368,970	\$7,038,127,887	\$7,536,366,110
ST Investment Returns	\$247,941,789	\$219,052,515	\$213,181,719	\$208,066,643	\$203,481,711	\$199,267,783	\$195,313,696	\$191,542,473	\$187,901,260	\$184,354,085
EBT = EBIT - Interest + ST Returns	\$131,472,387,460	\$143,923,405,149	\$156,519,958,229	\$169,412,976,414	\$182,573,684,006	\$196,421,451,823	\$210,597,371,983	\$224,988,965,875	\$239,752,913,638	\$254,995,325,146
Income Taxes = EBT * %EffTax (15.51%)	\$20,396,751,105	\$22,328,413,819	\$24,282,655,032	\$26,282,890,123	\$28,324,654,803	\$30,473,010,660	\$32,672,276,382	\$34,905,001,932	\$37,195,494,816	\$39,560,217,019
Net Income = EBT - Income Taxes	\$111,075,636,355	\$121,594,991,330	\$132,237,303,196	\$143,130,086,291	\$154,249,029,203	\$165,948,441,164	\$177,925,095,601	\$190,083,963,943	\$202,557,418,823	\$215,435,108,127
Dividends	\$15,525,131,801	\$15,879,492,946	\$16,271,553,727	\$16,701,630,303	\$17,170,510,589	\$17,678,889,409	\$18,228,773,075	\$18,821,035,846	\$19,456,196,296	\$20,135,262,019

FORECAST CASH BUDGET — Mean across Monte Carlo samples (USD)

Line Item	FY2025	FY2026	FY2027	FY2028	FY2029	FY2030	FY2031	FY2032	FY2033	FY2034
MODULE 1: Operating Activities										
Inflows from Sales	\$416,161,000,000	\$437,670,428,571	\$459,179,857,143	\$480,689,285,714	\$502,198,714,286	\$523,708,142,857	\$545,217,571,429	\$566,727,000,000	\$588,236,428,571	\$609,745,857,143
Total Inflows (Operating)	\$416,161,000,000	\$437,670,428,571	\$459,179,857,143	\$480,689,285,714	\$502,198,714,286	\$523,708,142,857	\$545,217,571,429	\$566,727,000,000	\$588,236,428,571	\$609,745,857,143
Payments for Purchases (COGS)	\$209,479,943,818	\$215,111,448,777	\$220,236,683,361	\$224,862,675,415	\$228,997,548,241	\$232,650,478,706	\$235,831,648,594	\$238,552,189,724	\$240,824,123,455	\$242,660,295,213
Operational Expenses (OpEx)	\$59,189,357,476	\$62,743,853,543	\$66,268,928,527	\$69,970,096,920	\$73,867,187,499	\$77,530,912,397	\$81,309,330,879	\$85,303,190,906	\$89,343,258,569	\$93,310,113,709
Income Tax	\$20,396,751,105	\$22,328,413,819	\$24,282,655,032	\$26,282,890,123	\$28,324,654,803	\$30,473,010,660	\$32,672,276,382	\$34,905,001,932	\$37,195,494,816	\$39,560,217,019
Total Outflows (Operating)	\$289,066,052,399	\$300,183,716,139	\$310,788,266,920	\$321,115,662,459	\$331,189,390,544	\$340,654,401,763	\$349,813,255,854	\$358,760,382,562	\$367,362,876,839	\$375,530,625,941
Operating Net Cash Balance	\$127,094,947,601	\$137,486,712,432	\$148,391,590,223	\$159,573,623,255	\$171,009,323,742	\$183,053,741,094	\$195,404,315,574	\$207,966,617,438	\$220,873,551,732	\$234,215,231,202

MODULE 2: Investing Activities										
Total Inflows (Investing)	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Investment in Fixed Assets (CapEx)	\$11,073,955,233	\$11,037,456,845	\$11,001,079,249	\$10,964,822,047	\$10,928,684,842	\$10,892,667,238	\$10,856,768,841	\$10,820,989,257	\$10,785,328,096	\$10,749,784,966
Total Outflows (Investing)	\$11,073,955,233	\$11,037,456,845	\$11,001,079,249	\$10,964,822,047	\$10,928,684,842	\$10,892,667,238	\$10,856,768,841	\$10,820,989,257	\$10,785,328,096	\$10,749,784,966
Investing Net Cash Balance	\$-11,073,955,233	\$-11,037,456,845	\$-11,001,079,249	\$-10,964,822,047	\$-10,928,684,842	\$-10,892,667,238	\$-10,856,768,841	\$-10,820,989,257	\$-10,785,328,096	\$-10,749,784,966

MODULE 3: External Financing										
ST Loan	\$87,204,056,424	\$96,117,022,855	\$105,620,475,218	\$115,735,721,110	\$126,483,234,187	\$137,882,470,325	\$149,951,679,605	\$162,707,716,402	\$176,165,850,080	\$190,339,579,048
LT Loan	\$17,210,551	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Total Inflows (Financing)	\$87,221,266,975	\$96,117,022,855	\$105,620,475,218	\$115,735,721,110	\$126,483,234,187	\$137,882,470,325	\$149,951,679,605	\$162,707,716,402	\$176,165,850,080	\$190,339,579,048
ST Principal Payment	\$88,271,000,000	\$87,204,056,424	\$96,117,022,855	\$105,620,475,218	\$115,735,721,110	\$126,483,234,187	\$137,882,470,325	\$149,951,679,605	\$162,707,716,402	\$176,165,850,080
ST Interest	\$3,509,926,558	\$3,467,501,599	\$3,821,908,568	\$4,199,795,075	\$4,602,008,375	\$5,029,362,563	\$5,482,631,266	\$5,962,540,163	\$6,469,759,434	\$7,004,896,238
LT Principal Payment	\$10,912,000,000	\$8,547,079,717	\$7,992,201,779	\$7,473,346,616	\$6,988,175,623	\$6,534,502,017	\$6,110,280,982	\$5,713,600,453	\$5,342,672,494	\$4,995,825,244
LT Interest	\$984,505,930	\$909,262,262	\$850,232,794	\$795,035,530	\$743,421,682	\$695,158,614	\$650,028,793	\$607,828,808	\$568,368,453	\$531,469,872
Total Outflows (Financing)	\$103,677,432,488	\$100,127,900,001	\$108,781,365,996	\$118,088,852,439	\$128,069,326,789	\$138,742,257,381	\$150,125,411,366	\$162,235,649,028	\$175,088,516,783	\$188,698,041,434
Financing Net Cash Balance	\$-16,456,165,513	\$-4,010,877,147	\$-3,160,890,779	\$-2,352,931,329	\$-1,586,092,602	\$-859,787,057	\$-173,731,761	\$472,067,375	\$1,077,333,297	\$1,641,537,614

MODULE 4: Transactions with Owners										
Invested Equity (Equity Financing)	\$17,790,069	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Total Inflows (Owners)	\$17,790,069	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Dividends from Last Year	\$15,625,131,801	\$15,879,492,946	\$16,271,553,727	\$16,701,630,303	\$17,170,510,589	\$17,678,889,409	\$18,228,773,075	\$18,821,035,846	\$19,456,196,296	\$20,135,262,019
Stock Buyback	\$81,829,331,551	\$105,661,644,567	\$116,519,131,174	\$127,854,196,266	\$139,406,897,100	\$151,520,333,873	\$163,881,694,680	\$176,389,602,716	\$189,171,897,156	\$202,314,177,972
Total Outflows (Owners)	\$97,354,463,352	\$121,541,137,514	\$132,790,684,901	\$144,555,826,569	\$156,577,407,689	\$169,199,223,282	\$182,110,467,755	\$195,210,638,562	\$208,628,093,452	\$222,449,439,991
Owners' Transaction Net Cash Balance	\$-97,336,673,283	\$-121,541,137,514	\$-132,790,684,901	\$-144,555,826,569	\$-156,577,407,689	\$-169,199,223,282	\$-182,110,467,755	\$-195,210,638,562	\$-208,628,093,452	\$-222,449,439,991

MODULE 5: Discretionary Transactions										
Return from ST Investments	\$247,941,789	\$219,052,515	\$213,181,719	\$208,066,643	\$203,481,711	\$199,267,783	\$195,313,696	\$191,542,473	\$187,901,260	\$184,354,085
Total Inflows (Discretionary)	\$247,941,789	\$219,052,515	\$213,181,719	\$208,066,643	\$203,481,711	\$199,267,783	\$195,313,696	\$191,542,473	\$187,901,260	\$184,354,085
Total Outflows (Discretionary)	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Discretionary Transaction Net Cash Balance	\$247,941,789	\$219,052,515	\$213,181,719	\$208,066,643	\$203,481,711	\$199,267,783	\$195,313,696	\$191,542,473	\$187,901,260	\$184,354,085

Reviewer's Feedback

(1) You identified the principle of “no plug” and some accounting identities, such as non-current assets (NCA) being a function of depreciation and capex:

$$NCA_t = NCA_t-1 - Depreciation_t + CapEx_t$$

However, later, while formulating the time-series model, you consider the non-current asset at time t to be driven by a ratio:

$$NCA_t = NCA_t-1 * (1 + \%AG)$$

Is it possible to design a workflow that fully respects the accounting identities (like the first equation) and can be driven by drivers?

Response: Thank you for catching this mistake. This was a typo in the report; the actual script had the correct relationship of $CapEx_t = \%AG * Sales_t$. We fixed the typo in the report

(2) What is the difference between the FIFO and LIFO principles in accounting? Under what circumstances will these two accounting choices make a difference? Is it possible to avoid using COGS as a plug?

Response: FIFO (First-In, First-Out): Under FIFO, the oldest inventory costs are matched against current sales revenues.

LIFO (Last-In, First-Out): Under LIFO, the most recent inventory costs are matched against current sales revenues.

The most difference can arise from these two accounting standards when there is a rapid inflationary (deflationary) environment. Under FIFO, the cost ratio will be smoother and slightly lower (higher) than the actual current cost of replacement. Under LIFO, the cost ratio will be more volatile since the recent inventory cost is used, and it will tightly track the inflation (deflation) and will squeeze (grow) the margin immediately. Under inflation = 0%, FIFO and LIFO should equal each other because old cost = new cost. Under high inflation, FIFO profit > LIFO profit.

In our model, we avoid using COGS as a plug by updating it so that we learn the cost ratio (CR_t) directly from the historical COGS and sales data by modeling cost ratio as a logit and learning the underlying parameters α and β to ensure positivity :

$$\text{logit}(CR_t) = \ln(CR_t / (1 - CR_t)) = \alpha + \beta \cdot t \text{ (from (5))}$$

$$CR_t = COGS_t / Sales_t \text{ (from (4))}$$

(3) Similar to (1), can you list out all the entities (rows) in the income statement, cash budget, and balance sheet (simplified versions are fine)? Then analyze how these three statements connect with each other. You may start with revenue on the income statement.

Response: We now include the 10-year forecasted Balance Sheet, Income Statement, and Cash Budget in **6. Projected Balance Sheet, Income Statement, and Cash Budget**.

(4) Advance payment is defined as: $AdvPP_t = \%AdvPP * Purchase_t + 1$

Is it possible to remove the future dependency on this variable? It may break causal timing and introduce leakage during training. Are there alternative approaches to model this term?

Response: We updated our model such that the advance payments for purchases (AdvPP) are proportional to this year's purchases instead of next year, with the justification:

- From 2.1.3. Advance Payments for Purchases : “Advance payments to suppliers for purchases (\$AdvPP\$) can be modeled as strictly proportional to the total purchases made during the current year. This approach relies on the simplifying assumption that current purchasing volumes serve as a reliable proxy for near-term future demand.”

Same is applied for advance payments from customers for sales (AdvPS):

- From 2.3.2. Advance Payments for Future Sales: “Advance payments received from customers for future sales (\$AdvPC\$) are modeled as a direct proportion of the current year's total sales. This assumes that the volume of upfront customer payments scales linearly with overall revenue generation.”

(5) For CapEx, it's great that you split this term into maintenance capex and growth capex:

$$CapEx_t = MaintenanceCapex_t + GrowthCapex_t$$

You further assume maintenance capex is tied to depreciation. This is a common approach when forecasting future cash flows by assuming a firm is in a steady state. What if the firm is not in a steady state?

Response: We updated our model to handle the case where the asset replacement rate is not unitary ($= 1.0$). We reflect that change in the report as well:

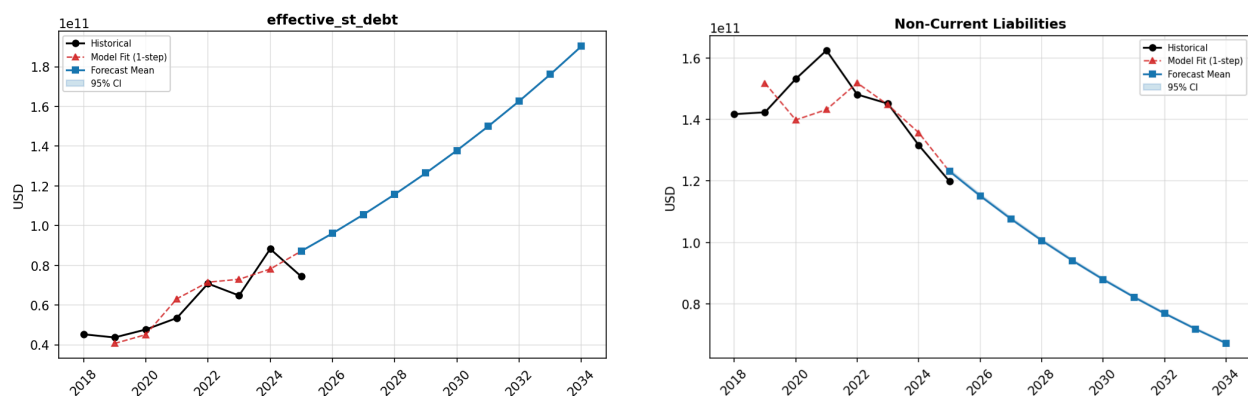
- From *Non-Current Assets (NCA)*: “To model CapEx, we diverge from a rigid steady-state assumption where maintenance perfectly offsets depreciation. Instead, we decompose CapEx into a maintenance component (scaled by a trainable parameter, %AM, to reflect partial, full, or inflationary replacement) and a growth component driven by sales expansion:

$$CapEx_t = \%AM \cdot Depreciation_t + \%AG \cdot Sales_t \quad (3)$$

(6) In your report, the financing dynamics on debt are as follows:

$$Debt_{t-1} \Rightarrow Interest_t, PrincipalDue_t \Rightarrow deficit_t \Rightarrow Debt_t$$

Can you plot the amount of debt and see whether the outcome is sensible?



Response: We have plotted the forecasted debt levels (see above). The outcome is sensible and mathematically stable. We verified that the model does not fall into an explosive 'debt spiral.' The financing dynamics converge realistically.

Specifically for Apple, the model accurately captures two key trends:

1. Short-Term Debt: The model predicts short-term debt scaling proportionally with Sales. This correctly reflects Apple's real-world strategy of utilizing revolving credit and commercial paper for working capital needs, despite possessing a large cash balance.
2. Long-Term Debt: The forecast accurately continues the historical trend of Apple slowly paying down its non-current liabilities. Ultimately, the plot confirms that the endogenous deficit calculation handles debt rollover gracefully without compounding errors.

(7) *Some rules seem a bit ad hoc. Is it possible to tie this to terms with a stronger financial relationship?*

Response: We can redefine our training variables from arbitrary percentages to these time-based efficiency metrics.

Receivables: Instead of %AR, we can train a target DSO (Days Sale Outstanding) parameter.

$$AR_t = DSO / 365 * Sales_t$$

Inventory: Inventory is a function of the cost to produce goods, not the final sales price. Tie it to COGS using DSI (Days Sales of Inventory)

$$Inv_t = DSI / 365 * COGS_t$$

Payables: We can use DPO (Days Purchases Outstanding) instead of %AP:

$$AP_t = DPO / 365 * Purchases_t$$

(8) Can you specify the possible constraints for the variables and equations? Please add them to the report and implement them in the code.

Response: Thank you. We have added a new section Parameter Constraints and Bounds

(9) *After you have trained the model, can you also perform inference and simulate a firm's financial statements for several years?*

Response: We now include the 10-year forecasted Balance Sheet, Income Statement, and Cash Budget in **6. Projected Balance Sheet, Income Statement, and Cash Budget.**

(10) Remember to write test cases.

Response: Thank you for the reminder. We wrote test cases for all the scripts appearing in this report:

simple_financial_model.py

trainable_financial_model.py

llm-only-balance-sheet-forecast.py

llm_extract_tax_anomalies_from_pdf.py

send_reasoning_prompt.py

llm_extract_financial_statement_from_pdf.py

llm_extract_json_from_financial_statements.py

The test scripts have 'test_' in front of each name.

Part 2

a) LLM Selection and Infrastructure Setup

My favorite LLM is DeepSeek V3.2 for reasoning which has 128k context window and gpt-5-nano for non-reasoning task which has 400k context window. From my daily usage, DeepSeekV3.2 follows instructions well with less hallucinations than other models from competitors. Gpt-5-nano, from my experience, is an inexpensive model that performs well for pretty complicated tasks but hallucinates more. I use Microsoft's Azure Foundry to access these models via Azure API. We have set up an AWS Lambda function that makes

the API calls so that we can hide the API keys for our Azure API. The AWS Lambda address is saved as an environment variable in a .env file, with its content being:

AZURE_DEEPSEEK_ENDPOINT=<https://dru54knwxa7sb4rlwrgws3kaze0jdvvtj.lambda-url.us-west-1.on.aws/>

We have included the .env file in the zipped file attached with our report submission.

b) LLM-Only Balance Sheet Forecasting

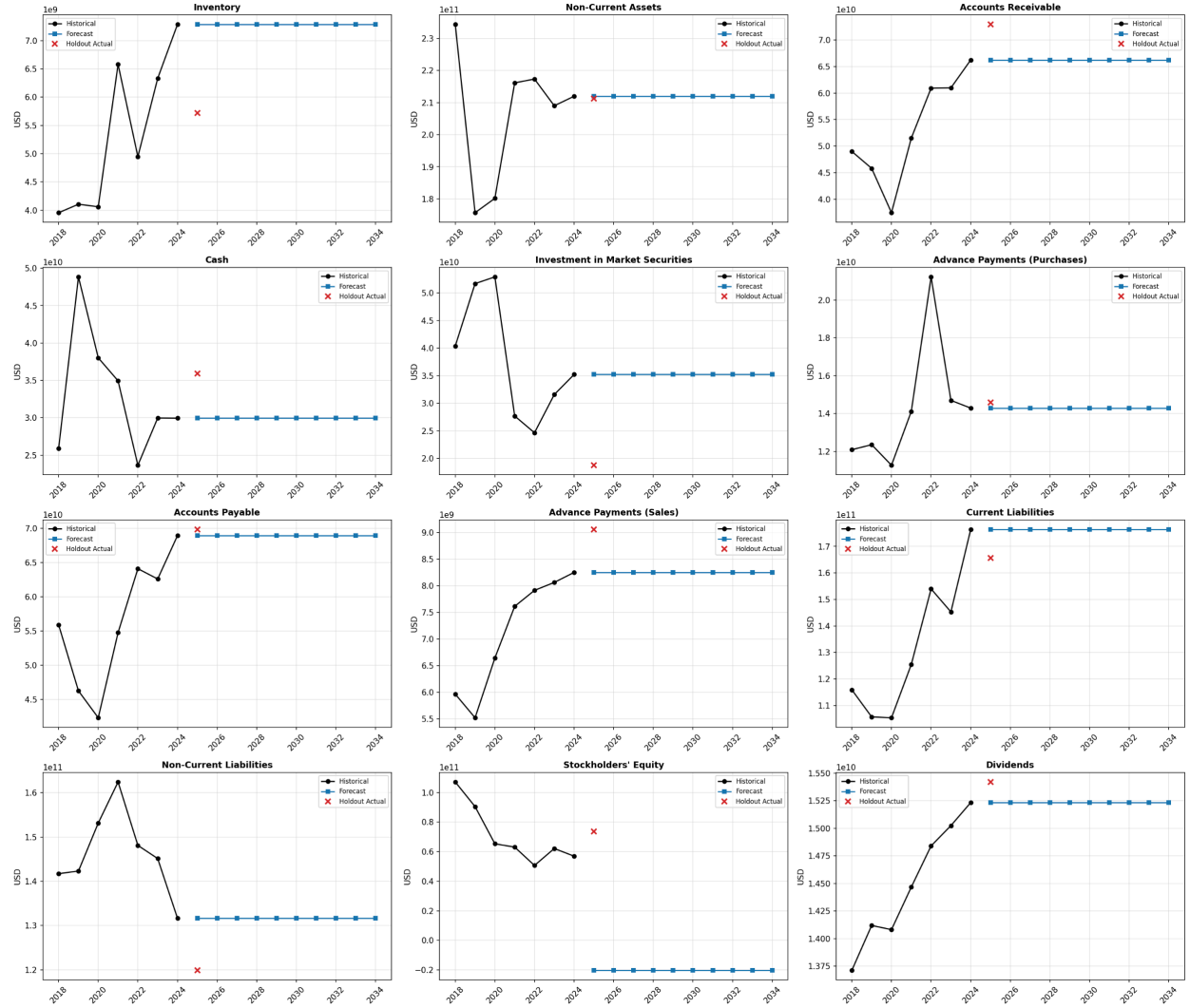
We can use the same input data from part 1 and feed it to DeepSeekV3.2 to forecast the balance sheet, as done in llm-only-balance-sheet-forecast.py.

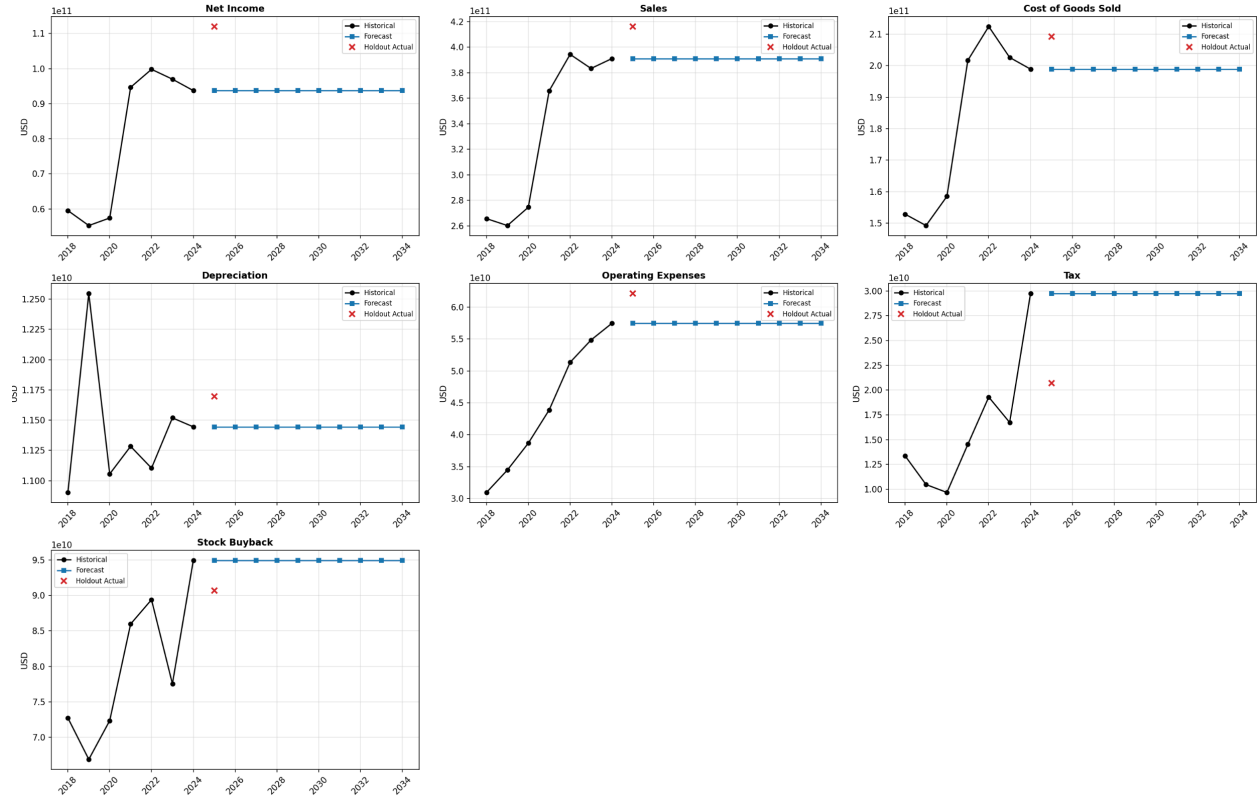
We use the following prompt with the identity check that assets = liabilities + equity:

```
prompt: Dict[str, Any] = {
    "task": "Forecast all required financial elements for future years. "
    "Return ONLY valid JSON (no markdown).",
    "historical_facts": historical_rows,
    "value_units": "trillions_of_usd",
    "value_scale_to_usd": LLM_AMOUNT_SCALE,
    "required_elements": ELEMENT_KEYS,
    "accounting_identity": (
        "inventory + nca + accounts_receivable + cash + "
        "investment_in_market_securities + advance_payments_purchases = "
        "accounts_payable + advance_payments_sales + current_liabilities + "
        "non_current_liabilities + equity"
    ),
    "output_schema": {
        "forecast": [
            {
                "inventory": "float",
                "nca": "float",
                "accounts_receivable": "float",
                "cash": "float",
                "investment_in_market_securities": "float",
                "advance_payments_purchases": "float",
                "accounts_payable": "float",
                "advance_payments_sales": "float",
                "current_liabilities": "float",
                "non_current_liabilities": "float",
                "equity": "float",
                "dividends": "float",
                "net_income": "float",
                "sales": "float",
                "cogs": "float",
                "depreciation": "float",
                "opex": "float",
                "tax": "float",
                "stock_buyback": "float",
            }
        ]
    }
}
```

The returned fields are further processed within the script **llm-only-balance-sheet-forecast.py** to ensure the balance sheet identity holds. The forecasted values are plotted below:

LLM Financial Forecast (blind, Identity-Constrained)



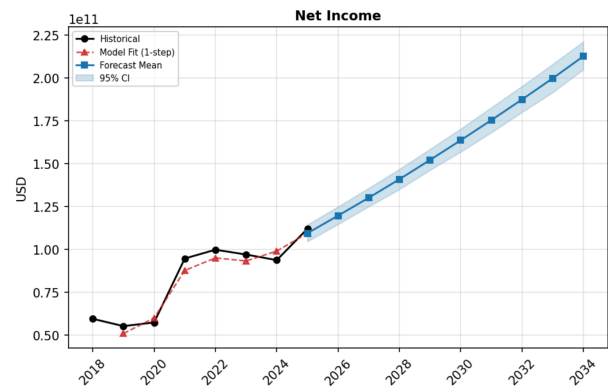
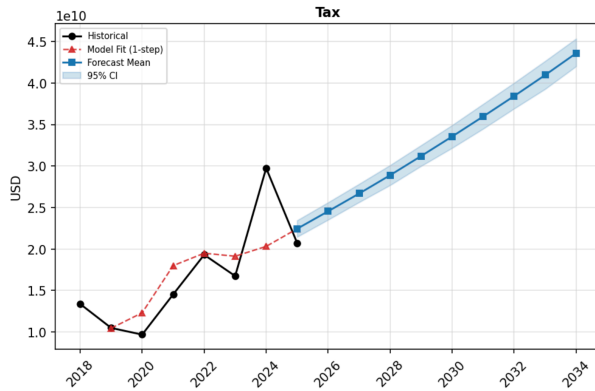


We see in “Stockholders’ Equity” graph that it drops to negative. This is because the LLM actually failed to return the projected balance sheet fields that abide by the accounting identity of assets = liabilities + equity, so the script’s procedure that ensures the identity results in negative equity value. It is clear that LLM-only forecasting of balance sheet fields is severely inadequate.

c) Combining LLM with the Balance Sheet Forecasting Model (Tax Anomalies)

Is it possible to combine LLM with the balance sheet forecasting model? Yes. When we look at the forecasted model, we see that one of the major contributors for predicted income discrepancy from data is the calculated tax (income underestimated by ~5 billion, and tax overestimated by ~2 billion). Net income (~100 billion dollars) is sales – cogs – opex – depreciation – debt interest + market securities return – tax, and we actually see good predicted values for COGS, opex, and depreciation (market securities return and interest payments are uncertain due to incomplete data, but their combined amount is order of magnitude smaller than tax) all show discrepancy from data of less than 1 billion, while tax, which takes ~30% of net income, is off from the data by ~2 billion, almost 10% higher than predicted which lowers the predicted net income.

In the case of Apple, there was a spike in effective tax rate in 2024 due to a one-time income tax charge resulting from the final resolution of the European Commission State Aid Decision. This skewed the learned effective tax rate upwards during training, as evidenced by the consistent over-estimation of tax for one-step forecast model fit.



To avoid this issue, we can ask LLM to notify us of any one-time tax payments like this so that we can appropriately handle it during training. When calculating the MSE between predicted and actual tax owed, the ground truth data can be offsetted by the one-time amount to better learn the baseline effective tax rate.

We can feed the 10-K to LLM and ask it to identify pages that contain the one-off tax information, and extract this data in JSON format for model consumption:

Script: `llm_extract_tax_anomalies_from_pdf.py`

```
DEFAULT_SELECTION_QUERY = (
    "Item 7 Management's Discussion and Analysis (MD&A), "
    "and Item 8 Financial "
    "Statements and Supplementary Data sections specifically "
    "covering "Income Taxes" "
    "and "Commitments and Contingencies".'"
)
```

```
DEFAULT_SELECTION_PROMPT = (
    "You are given pages from a Form 10-K annual report.\n"
    "Identify all pages that contain any of the following:\n"
    "1) Item 7: Management's Discussion and Analysis (MD&A)\n"
    "2) Item 8: Financial Statements and Supplementary Data notes "
    "for:\n"
    " - Income Taxes\n"
    " - Commitments and Contingencies\n\n"
    "Include pages with headings, substantive discussion, tables, "
    "and continuations of "
    "these sections.\n"
    "Return ONLY valid JSON with this exact shape: '"
    '{"pages": [<page_number>, ...]}'.\n'
    "If none match, return {'pages': []}'.\n\n"
    "Query: {query}\n\n"
    "Pages:\n{pages}"
)
```

```
DEFAULT_EXTRACTION_PROMPT = (
    "You are an expert financial analyst. Your task is to analyze the "
    "provided excerpts "
    "from a company's Form 10-K (MD&A and Financial Footnotes) and "
    "extract data regarding "
    "one-time tax anomalies and future tax contingencies.\n\n"
    "Carefully evaluate the text for the following:\n\n"
    "Current Year Anomalies: Identify any massive, non-recurring, "
    "discrete tax charges "
```

```

"or benefits that heavily skewed the current year's net income "
"(e.g., finalized state "
"aid decisions, sudden impacts from new tax legislation).\n\n"
"Future Contingencies: Identify any quantified maximum tax "
"exposures, unreserved tax "
"liabilities, or significant unrecognized tax benefits (UTBs) "
"that management indicates "
"could be resolved or assessed in future years (e.g., tax funds "
"held in escrow pending "
"appeal, estimated UTB decreases in the next 12 months).\n\n"
"You must respond ONLY with a valid JSON object using the exact "
"schema below. Do not "
"include any markdown formatting, preamble, or postscript. If a "
"specific data point is "
"not explicitly mentioned or cannot be reliably quantified from "
"the text, output null "
"for that field.\n\n"
"JSON Schema:\n"
"{{\n"
'  "tax_onetime_amount": <number in billions or null>,\n'
'  "tax_onetime_note": "<string explaining the anomaly or null>\",\n'
'  "tax_contingency_amount": <number in billions or null>,\n'
'  "tax_contingency_note": "<string explaining the future reserve '
'or null>"\n'
"}}\n\n"
"Pages:\n{pages}"
)

```

LLM was able to extract the one-time tax payment from this portion of 10-K:

2024 10-K page 24:

Provision for Income Taxes

Provision for income taxes, effective tax rate and statutory federal income tax rate for 2024, 2023 and 2022 were as follows (dollars in millions):

	2024	2023	2022
Provision for income taxes	\$ 29,749	\$ 16,741	\$ 19,300
Effective tax rate	24.1%	14.7%	16.2%
Statutory federal income tax rate	21%	21%	21%

The Company’s effective tax rate for 2024 was higher than the statutory federal income tax rate due primarily to a one-time income tax charge of \$10.2 billion, net, related to the State Aid Decision (refer to Note 7, “Income Taxes” in the Notes to Consolidated Financial Statements in Part II, Item 8 of this Form 10-K) and state income taxes, partially offset by a lower effective tax rate on foreign earnings, the impact of the U.S. federal R&D credit, and tax benefits from share-based compensation.

The Company’s effective tax rate for 2024 was higher compared to 2023 due primarily to a one-time income tax charge of \$10.2 billion, net, related to the State Aid Decision, a higher effective tax rate on foreign earnings and lower tax benefits from share-based compensation.

And page 39:

Note 7 – Income Taxes

European Commission State Aid Decision

On August 30, 2016, the Commission announced its decision that Ireland granted state aid to the Company by providing tax opinions in 1991 and 2007 concerning the tax allocation of profits of the Irish branches of two subsidiaries of the Company (the “State Aid Decision”). The State Aid Decision ordered Ireland to calculate and recover additional taxes from the Company for the period June 2003 through December 2014. Irish legislative changes, effective as of January 2015, eliminated the application of the tax opinions from that date forward. The recovery amount was calculated to be €13.1 billion, plus interest of €1.2 billion.

From time to time, the Company requested approval from the Irish Minister for Finance to reduce the recovery amount for certain taxes paid to other countries. As of September 28, 2024, the adjusted recovery amount of €12.7 billion plus interest of €1.2 billion was held in escrow and restricted from general use. The total balance of the escrow, including net unrealized investment gains, was €14.2 billion or \$15.8 billion as of September 28, 2024, of which \$2.6 billion was classified as cash and cash equivalents and \$13.2 billion was classified as current marketable securities in the Consolidated Balance Sheet. Refer to the Cash, Cash Equivalents and Marketable Securities section of Note 4, “Financial Instruments” for more information.

Apple Inc. | 2024 Form 10-K | 39

The Company and Ireland appealed the State Aid Decision to the General Court of the Court of Justice of the European Union (the “General Court”). On July 15, 2020, the General Court annulled the State Aid Decision. On September 25, 2020, the Commission appealed the General Court’s decision to the European Court of Justice (the “ECJ”) and a hearing was held on May 23, 2023. On September 10, 2024, the ECJ announced that it had set aside the 2020 judgment of the General Court and confirmed the Commission’s 2016 State Aid Decision. As a result, during the fourth quarter of 2024 the Company recorded a one-time income tax charge of \$10.2 billion, net, which represents \$15.8 billion payable to Ireland via release of the escrow, partially offset by a U.S. foreign tax credit of \$4.8 billion and a decrease in unrecognized tax benefits of \$823 million.

JSON output:

```
{
  "query": "Item 7 Management's Discussion and Analysis (MD&A), and Item 8 Financial Statements and Supplementary Data sections specifically covering \"Income Taxes\" and \"Commitments and Contingencies\".",
  "selected_pages": [
    23, 24, 25, 26, 27, 41, 42, 43, 45, 46, 47
  ],
  "extraction": {
    "tax_onetime_amount": 10.2,
    "tax_onetime_note": "One-time income tax charge of $10.2 billion net related to the European Commission State Aid Decision.",
    "tax_contingency_amount": 13,
    "tax_contingency_note": "Gross unrecognized tax benefits could decrease between approximately $5 billion and $13 billion in the next 12 months, primarily related to intercompany transfer pricing and deemed repatriation tax."
  }
}
```

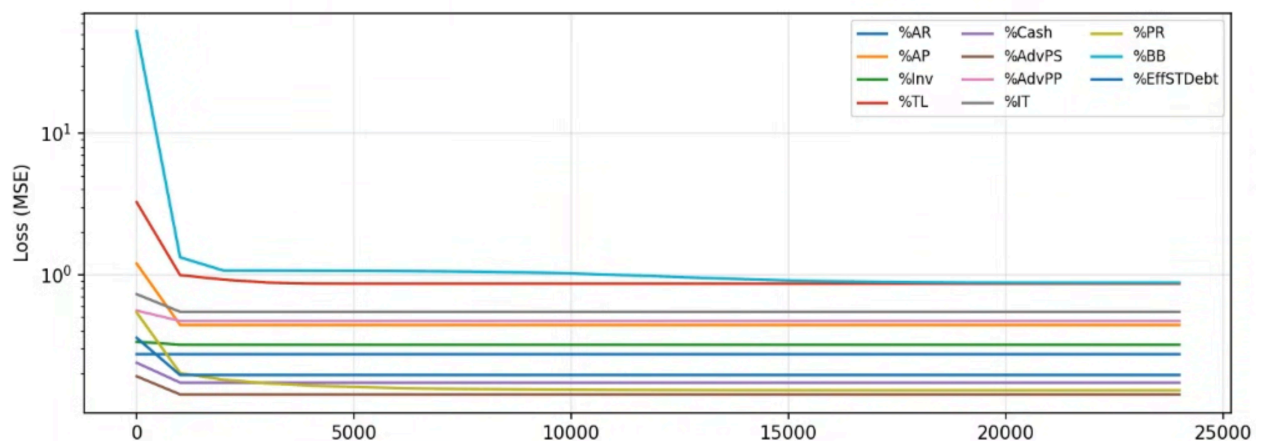
We update our balance sheet forecasting script to include the `tax_onetime_amount` for better learning of the baseline tax rate. We run the script `trainable_financial_model_taxanomaly_effstdebt_bayesian-vi_refactored.py` with `include_tax_anomalies=True`.

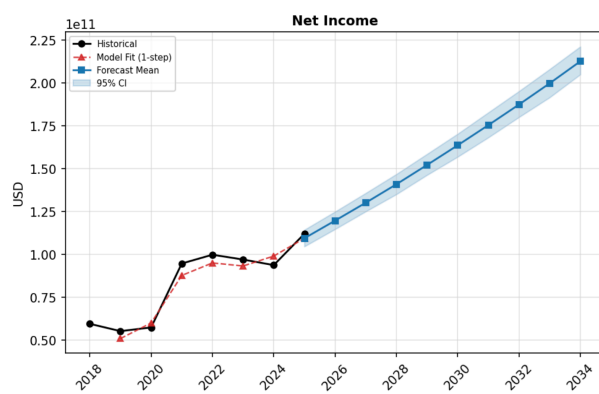
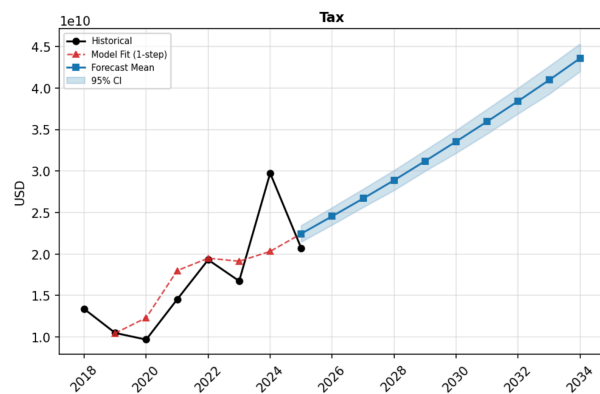
Effective tax rate drops from 17% to 15.5%! And the learning loss is smaller for the effective tax rate %IT:

Before:

```
-----
Simple parameters:
Final %AG: 0.00000
Final %AM: 0.94062
Final %Depr: 0.05553
Final %AdvPS: 0.02126
Final %AdvPP: 0.07840
Final %AR: 0.15866
Final %AP: 0.30794
Final %Inv: 0.01603
Total Liquidity (baseline + logit-linear): baseline=0.6013, alpha=-2.3463, beta=-0.411579
=> %TL at t=0: 0.0874, %TL at t=7: 0.0053
Cash % of Liquidity (logit-linear): alpha=-0.2235, beta=0.032385
=> %Cash at t=0: 0.4444, %Cash at t=7: 0.5008
Final %IT: 0.17021
Final %PR: 0.99969
Final DivAdjSpeed ( $\alpha$ ): 0.00371
Stock Buyback (baseline + ratio*depr): baseline=1.5540, ratio=-6.610795
Effective ST Debt % of Sales (logit-linear): alpha=-1.7458, beta=0.059738
=> %EffSTDebt at t=0: 0.1486, %EffSTDebt at t=7: 0.2095
Cost Ratio (logit-linear): alpha=0.3459, beta=-0.0475
=> CR at t=0: 0.5856, CR at t=7: 0.5034
Bayesian OpEx Variable %: Mean=0.0923, Std=0.0155
Bayesian OpEx Baseline (USD): Mean=3.91e+10, Std=8.09e+08
OpEx aleatoric uncertainty (USD): 2.49e+09
-----

Structural parameters:
Final %AvgSTInt: 0.03970
Final %AvgLTInt: 0.00692
Final AvgM: 15.36485
Final %MSReturn: 0.01119
Equity Financing % (logit-linear): alpha=0.5389, beta=0.018432
=> %EF at t=0: 0.6316, %EF at t=7: 0.6610
-----
```





Before accounting for tax anomalies. The predicted net income is 1.09e11 dollars, which is off by 2.68%, or 3 billion dollars, from the actual value of 1.12e11 dollars.

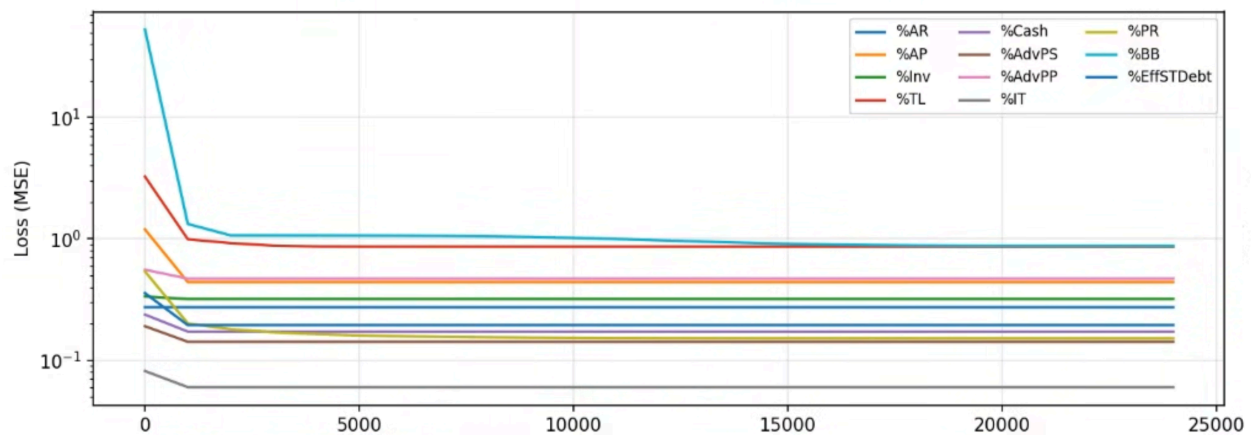
After:

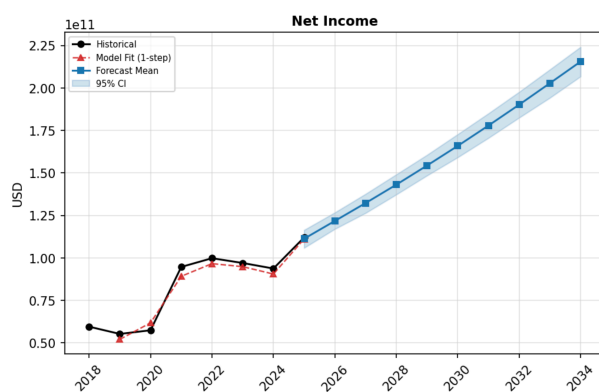
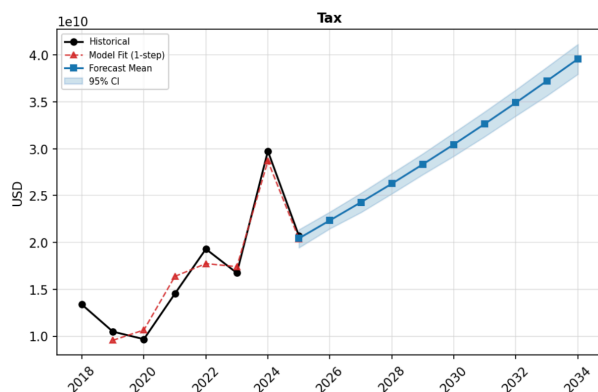
```

-----
Simple parameters:
Final %AG: 0.00000
Final %AM: 0.94062
Final %Depr: 0.05553
Final %AdvPS: 0.02126
Final %AdvPP: 0.07840
Final %AR: 0.15866
Final %AP: 0.30794
Final %Inv: 0.01603
Total Liquidity (baseline + logit-linear): baseline=0.6013, alpha=-2.3463, beta=-0.411579
=> %TL at t=0: 0.0874, %TL at t=7: 0.0053
Cash % of Liquidity (logit-linear): alpha=-0.2235, beta=0.032385
=> %Cash at t=0: 0.4444, %Cash at t=7: 0.5008
Final %IT: 0.15514
Final %PR: 0.99969
Final DivAdjSpeed ( $\alpha$ ): 0.00371
Stock Buyback (baseline + ratio*depr): baseline=1.5540, ratio=-6.610795
Effective ST Debt % of Sales (logit-linear): alpha=-1.7458, beta=0.059738
=> %EffSTDebt at t=0: 0.1486, %EffSTDebt at t=7: 0.2095
Cost Ratio (logit-linear): alpha=0.3459, beta=-0.0475
=> CR at t=0: 0.5856, CR at t=7: 0.5034
Bayesian OpEx Variable %: Mean=0.0957, Std=0.0160
Bayesian OpEx Baseline (USD): Mean=3.90e+10, Std=8.05e+08
OpEx aleatoric uncertainty (USD): 2.45e+09
-----

Structural parameters:
Final %AvgSTInt: 0.03976
Final %AvgLTIInt: 0.00691
Final AvgM: 15.40353
Final %MSReturn: 0.00704
Equity Financing % (logit-linear): alpha=0.6014, beta=-0.081179
=> %EF at t=0: 0.6460, %EF at t=7: 0.5083
-----

```





After accounting for tax anomalies. The predicted net income in 2025 is 1.11e11 dollars, which is only off by 0.89%, or 1 billion dollars, from the actual value of 1.12e11 dollars! The income tax MSE loss during training also drops by an order of magnitude.

d) Strategic Recommendations for CEO/CFO

For recommendation to CEO/CFO given our result, we feed the following prompt, and attach the historical and 10-year forecast of balance sheet and income statement:

Prompt = “ ”

You are an elite Corporate Finance Advisor and Strategic Capital Expert at a top-tier investment bank. You specialize in advising the CEOs and CFOs of mature, cash-generative mega-cap technology firms on capital structure optimization, liquidity management, and capital allocation.

Context & Task:

I will provide you with two sets of financial data for a target company:

Historical Baseline: Actual financial performance from recent years.

ML Forecast: A multi-year financial forecast (Balance Sheet, Income Statement, and Cash Budget) generated by an advanced Monte Carlo machine learning ensemble.

Your task is to analyze the historical data against the forecast to identify the three most critical financial leverage points (risks or opportunities) over the forecasted period.

STRICT GUARDRAILS (CRITICAL):

NO OPERATIONAL ADVICE: You are strictly forbidden from recommending operational, marketing, or business strategy changes. Do not suggest “launching new products,” “increasing sales efforts,” or “improving supply chain efficiency.”

STRICTLY FINANCIAL DOMAIN: Your recommendations must be confined purely to Corporate Finance: Capital Structure (debt/equity mix, WACC), Liquidity & Treasury (working capital, commercial paper), and Capital Allocation (dividends, share repurchases, CapEx).

MANDATORY EVIDENCE: You cannot make a claim without citing specific numbers from the provided tables.

Analytical Heuristics (The Mega-Cap Tech Lens & Trend Analysis):

Historical Benchmarking (Crucial): Always contextualize the forecast by comparing it to the historical baseline. Identify structural breaks (where the forecast deviates significantly from historical trends) or trend accelerations.

Strategic Leverage vs. Distress: Do not penalize shrinking Accounting Equity. If Equity is declining due to aggressive Stock Buybacks and Dividends, treat this as highly efficient Capital Return and WACC optimization—NOT financial distress—provided Net Income and Free Cash Flow remain historically strong.

Net Debt Focus: Evaluate liquidity by looking at Cash + Marketable Securities against Liabilities. Mega-cap tech companies often issue debt simply to lower their cost of capital; evaluate if the forecasted debt load is easily serviced by operating cash flows relative to historical interest coverage.

Operating Leverage: Note if the forecast predicts Revenue and Net Income growing at a faster or slower rate than historical Operating Expenses (OpEx).

Output Format:

Provide exactly 3 strategic priorities. For each, use this exact structure:

Observation: [State the specific quantitative trend, explicitly contrasting the forecasted figures against the historical baseline. Cite exact numbers and years].

Implication: [Explain the financial/credit risk or the WACC optimization opportunity this trend creates for the balance sheet].

Recommendation: [State the precise corporate finance action the CFO/CEO must execute to manage this specific trajectory].

Data Input:

[Insert Historical Tables / JSON here]

[Insert ML Forecast Tables / JSON here]

“ ” ”

The prompt with the context of the 10-year forecast is sent to our reasoning model by calling `send_reasoning_prompt.py`

And here is the recommendation that our reasoning model generated:

Analysis of Financial Leverage Points

Observation: The forecast projects a significant and accelerating decline in Stockholders' Equity, from \$70.7B in FY2025 to \$40.8B in FY2034, while Net Income grows from \$111.1B to \$215.4B over the same period. This is a structural break from the historical baseline (2017-2023), where Equity averaged \$78.5B and showed no consistent downward trend, even as Net Income grew.

Implication: This trajectory indicates an extremely aggressive capital return policy is embedded in the forecast, far exceeding historical levels. While this optimizes WACC by shrinking the equity base, it creates a critical dependency on continuous, high levels of free cash flow. The balance sheet is being deliberately levered, shifting from an equity-heavy to a more debt-dependent structure.

Recommendation: The CFO must formally establish and communicate a **Sustainable Capital Return Framework**. This framework should define the maximum allowable annual capital return (dividends + buybacks) as a percentage of forward-looking Free Cash Flow (e.g., 80-90%), ensuring it does not jeopardize the credit rating or the ability to service the forecasted rising debt load. This moves the policy from implicit to explicit, managing investor expectations and credit agency concerns.

Observation: The forecast shows a dramatic increase in **Effective Short-Term Debt**, which balloons from \$87.2B in FY2025 to \$190.3B by FY2034. Concurrently, highly liquid assets (Cash + Marketable Securities) are projected to decline from a combined \$62.3B to \$60.2B. This creates a rising **Net Debt** position. Historically (2020-2023), the company maintained a robust liquidity buffer, with Cash + Securities averaging \$55.8B against much lower debt levels.

Implication: The company is forecasted to become significantly more reliant on short-term wholesale funding. This introduces substantial refinancing and interest rate risk, especially as interest payments are projected to rise from \$4.5B to \$7.5B. The shrinking liquidity coverage ratio could pressure credit spreads and commercial paper rates during market stress.

Recommendation: Execute a **Strategic Liability Management Exercise**. The Treasury must term out a significant portion of the forecasted short-term debt into longer-dated fixed-rate bonds. This locks in historically low(er) long-term rates for a mega-cap tech issuer, reduces refinancing risk, and better matches the duration of the company's long-lived assets. Concurrently, establish a minimum liquidity covenant (e.g., $\text{Cash} + \text{Securities} \geq 1.0 \times \text{ST Debt}$) for internal risk management.

Observation: The forecast predicts a powerful **positive operating leverage** trend. From FY2025 to FY2034, Revenue grows by 46.5% (\$416B to \$610B), while the combined growth of COGS and OpEx is only 39.5%. This divergence accelerates EBIT margin expansion. This is a continuation but acceleration of a historical trend; from 2017-2023, Revenue grew 47.2% while COGS+OpEx grew 45.8%.

Implication: This trajectory will generate super-linear growth in Free Cash Flow, the fundamental driver for all capital allocation decisions (debt service, buybacks, dividends). It provides immense balance sheet flexibility and debt capacity. However, it also raises the strategic opportunity cost of holding excess, low-yielding cash on the balance sheet.

Recommendation: Initiate a **Dynamic Share Repurchase Program** tied to Free Cash Flow Surplus. Instead of a fixed buyback amount, the program should automatically allocate a percentage of quarterly FCF in excess of a pre-determined threshold (e.g., FCF above levels needed for debt service and a baseline dividend) to repurchases. This mechanism directly links the benefit of operating leverage to per-share earnings accretion and WACC optimization, while maintaining a cash safety net.

e, f, g) Financial Ratios and Automated Extraction Pipeline

We need to calculate cost-to-income, quick ratio, debt-to-equity ratio, debt-to-assets ratio, debt-to-capital ratio, debt-to-EBITDA ratio, and interest coverage ratio:

1. Cost-to-Income ratio = Operating Expense / Revenue

= Describes company's operating efficiency. Operating Expense includes cost of revenue.

- LLM to extract **Revenue** and **Operating Expense** from **Income Statement**. Operating Expense may not be a simple line item, may need to combine a few lines, e.g., GM has line for "Automotive and other cost of sales", "GM Financial interest, operating and other expenses" and "Automotive and other selling, general and administrative expense"

2. Quick Ratio = (Cash & Cash Equivalents + Short-Term Market Securities + Accounts Receivable) / Current Liabilities

= Ability to meet short-term liabilities with the most liquid assets

- LLM to extract **Cash & Cash Equivalents**, **Short-Term Market Securities**, **Accounts Receivable**, and **Current Liabilities** from **Balance Sheet**

3. Debt-to-Equity ratio = (Short-Term Debt + Long-Term Debt) / Total Equity

- LLM to extract **Total Debt** (including short-term and long-term debts) and **Total Equity** from **Balance Sheet**

4. Debt-to-Assets ratio = (Short-Term Debt + Long-Term Debt) / Total Assets

- Total Debt already extracted. LLM to extract **Total Assets** from **Balance Sheet**

5. Debt-to-Capital ratio = (Short-Term Debt + Long-Term Debt) / (Short-Term Debt + Long-Term Debt + Total Equity)

- Total Debt and Total Equity already extracted.

6. Debt-to-EBITDA ratio = (Short-Term Debt + Long-Term Debt) / EBITDA

= How many years it would take for a company to pay back its debt using its EBITDA

- Total Debt already extracted. LLM to extract **Net Income**, **Taxes**, and **Interest Expenses** from **Income Statement**; and **Depreciation and Amortization** from **Cash Flows** to calculate EBITDA

$$= \text{Net Income} + \text{Interest Expenses} + \text{Taxes} + \text{Depreciation} + \text{Amortization}$$

7. Interest Coverage ratio = EBIT / Interest Expense

= How easily a company can pay interest on its outstanding debt

- Interest Expense already extracted. Calculate EBIT = Net Income + Interest Expenses + Taxes

List of fields to be extracted:

Balance Sheet

- Cash & Cash Equivalents
- Short-Term Market Securities
- Accounts Receivables
- Total Current Liabilities
- Total Debt (Short-Term and Long-Term)
- Total Equity
- Total Assets

Income Statement

- Revenue
- Operating Expenses
- Net Income
- Taxes
- Interest Expenses

Cash Flows

- Depreciation and Amortization

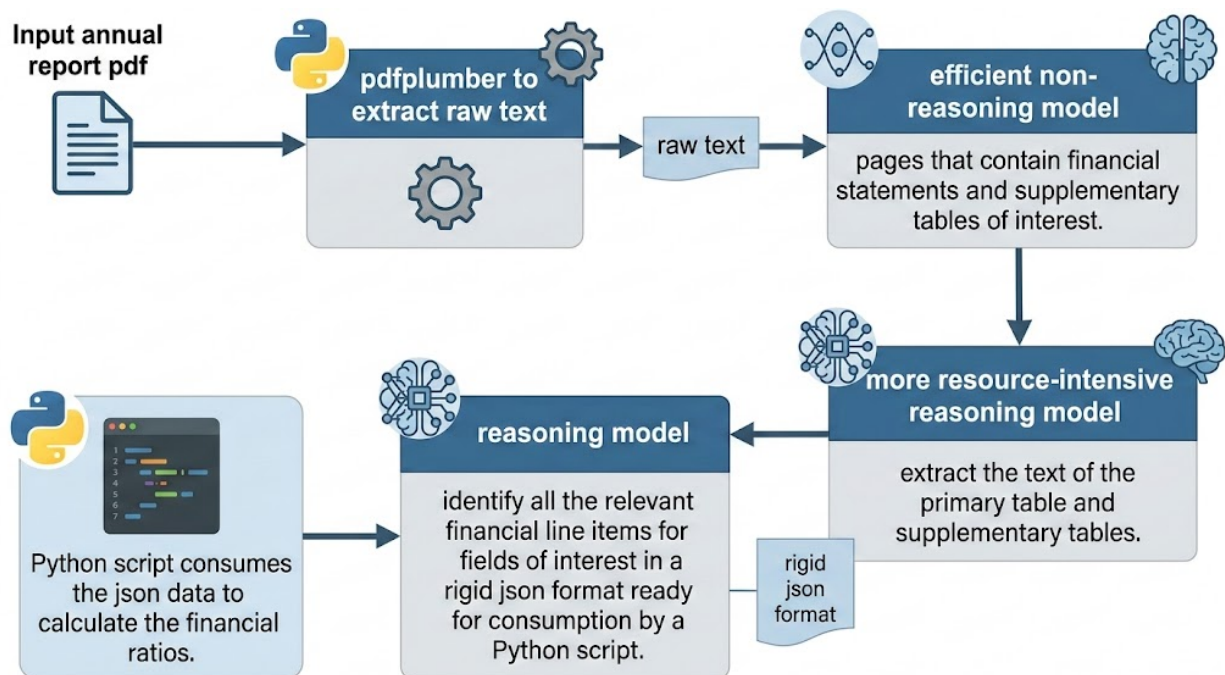
When extracting these values, also give LLM context in which we are obtaining these values, such that in case there isn't a 1:1 mapping of these variables to line items in these statements, LLM can assign the relevant fields and/or combine fields to return the field of interest (e.g., operating expenses in ExxonMobile include:

- Crude oil and product purchases
- Production and manufacturing expenses
- Selling, general and administrative expenses
- Depreciation and depletion
- Exploration expenses, including dry holes
- Non-service pension and postretirement benefit expense
- Other taxes and duties

When extracting the financial statements, also we had to make sure that we extract any supplementary tables. For example, Google's Income Statement had a line for 'Other Income' which combined interest income, interest expenses, and others, without listing out individual components. Only in a separate 'Other Income (Expenses), Net' table do we see the line item 'Interest expense' which we need for calculating

Interest Coverage ratio and Debt-to-EBITDA ratio. So, we had to update the LLM query to also extract supplementary tables.

The extraction step is broken down into three LLM agents for extraction and a Python script for actual calculation:



Input annual report pdf -> pdfplumber to extract raw text -> Feed raw text to efficient non-reasoning model to identify pages that contain financial statements and supplementary tables of interest -> Feed the relevant pages to the more resource-intensive reasoning model to extract the text of the primary table and supplementary tables -> Feed the extracted text data to the reasoning model to identify all the relevant financial line items for fields of interest in a rigid json format ready for consumption by a Python script -> Python script consumes the json data to calculate the financial ratios

We have tried using embeddings as done in [Zhang(25)], but we noticed that it is not effective in 2 ways:

1. Vectorization of each page fails to encapsulate all the semantic content. For example, if a page contains a supplementary table of interest along with other supplementary tables and texts, the vectorization of that page may not align closely with our query vector for that supplementary table of interest, and we cannot retrieve that information.
2. Embeddings are more sensitive than LLM to use of different synonyms or languages. We have observed from our attempt to use embedding models on Alibaba's annual report that our query for 'Consolidated Income Statement' failed to retrieve the page with "合并利润表"

LLMs excel in this area because they are incredibly good at semantic understanding and most of them are trained on multi-lingual data, giving them multi-lingual ability as an emergent property. However, LLMs are susceptible to hallucination. If you try to get it to do too many things at once, or perform a highly quantitative task, it will hallucinate and produce wrong answers. It is essential to break the task down into a step-by-step process that reduces hallucination and response quality, similar to how chain-of-thought prompting improves output quality [Wei(22)]. Also, by first extracting the relevant text before feeding it to the LLM to extract the financial fields, we prevent context dilution which is known to degrade output quality [Liu(23)].

llm_extract_financial_statement_from_pdf.py

Primary financial statement extraction:

Step 1: Page number identifying step

```
prompt = (
    "You are given multiple pages from an annual report. "
    f"Identify which pages contain the table "
    "for the requested query. "
    'Return ONLY JSON in the shape: {"pages": '
    ' [<page_number>, ...]}. '
    f"Include all pages that contain the table or line items. "
    'If none, return {"pages": []}. \n\n'
    f"Query: {query}\n\n"
    f"Pages: \n{pages_text}"
)
```

Where:

- **query** is one of ["Consolidated Balance Sheet", "Consolidated Income Statement", "Consolidated Cash Flow Statement"]
- **pages_text** is the entire text from annual report directly extracted with pdfplumber

Step 2: Table extraction step

```
prompt = (
    f"Find the table that corresponds to {query} and output it "
    f"in a nice tabular form from the following pages data.\n"
    f"Pages: \n{pages_text}"
)
```

Where:

- **query** is one of ["Consolidated Balance Sheet", "Consolidated Income Statement", "Consolidated Cash Flow Statement"]
- **pages_text** is the portion of the pdfplumber-extracted text from only the pages identified in Step 1

Output from Step 2:

Alphabet Inc. Consolidated Statements of Income
(in millions, except per share amounts)

Line Item	2022	2023	2024
Revenues	\$282,836	\$307,394	\$350,018
Costs and expenses:			
Cost of revenues	126,203	133,332	146,306
Research and development	39,500	45,427	49,326
Sales and marketing	26,567	27,917	27,808
General and administrative	15,724	16,425	14,188
Total costs and expenses	207,994	223,101	237,628
Income from operations	74,842	84,293	112,390
Other income (expense), net	(3,514)	1,424	7,425
Income before income taxes	71,328	85,717	119,815
Provision for income taxes	11,356	11,922	19,697
Net income	\$59,972	\$73,795	\$100,118
Basic net income per share (Note 12)	\$4.59	\$5.84	\$8.13
Diluted net income per share (Note 12)	\$4.56	\$5.80	\$8.04

Source: Alphabet 2024 Annual Report, page 57

Supplementary tables extraction:

Step 3: Page number identifying step

```

supplementary_query = (
    f"Supplementary disclosures, breakdowns, expansions, "
    f"and note tables for {query}. "
    f"Use this extracted primary statement as context to "
    f"infer relevant line items and "
    f"find related supplementary pages: {primary_context}"
)

prompt = (
    "You are given multiple pages from an annual report. "
    f"Identify which pages contain the table "
    "for the requested query. "
    'Return ONLY JSON in the shape: {"pages": '
    ' [<page_number>, ...]}. '
    f"Include all pages that contain the table or line items. "
    'If none, return {"pages": []}. \n\n'
    f"Query: {supplementary_query}\n\n"
    f"Pages: \n{pages_text}"
)

```

Where:

- **query** is one of ["Consolidated Balance Sheet", "Consolidated Income Statement", "Consolidated Cash Flow Statement"]
- **primary_context** is the extracted table from Step 2

- `pages_text` is the entire text from annual report directly extracted with pdfplumber

Step 4: Table extraction step

```
prompt = (
    "You are extracting supplementary financial disclosures "
    "from annual report pages.\n"
    f"Primary statement: {query}\n"
    "Primary statement extraction context:\n"
    f"{primary_context}\n\n"
    "Infer the line items from the primary statement context "
    "above. "
    "Extract supplementary tables related to those inferred "
    "line items "
    "(for example, breakdowns/expansions/schedules/notes such "
    "as an expanded "
    "'Other Income' table). "
    "You can be generous with the supplementary tables you "
    "extract since it is better to have more than not have "
    "necessary information.\n"
    "Return in a nice tabular format.\n"
    'If none are found, return "No supplementary tables "'
    'found'.\n\n'
    f"Pages:\n{pages_text}"
)
```

Where:

- `query` is one of ["Consolidated Balance Sheet", "Consolidated Income Statement", "Consolidated Cash Flow Statement"]
- `primary_context` is the extracted table from Step 2
- `pages_text` is the portion of the pdfplumber-extracted text from only the pages identified in Step 3

Output from Step 4:

Based on the primary Consolidated Income Statement context, the inferred line items are: Revenues, Cost of revenues, Research and development, Sales and marketing, General and administrative, Income from operations, Other income (expense), net, Income before income taxes, Provision for income taxes, and Net income.

The provided notes to the financial statements contain several supplementary tables that directly expand upon these line items, particularly **Revenues**, **Other income (expense), net**, and **Provision for income taxes**.

Here are the extracted supplementary tables:

1. Disaggregated Revenues (Note 2, Page 69)

This table provides a detailed breakdown of the "Revenues" line item.

By Type

Revenue Type	2022	2023	2024
Google Search & other	\$162,450	\$175,033	\$198,084
YouTube ads	29,243	31,510	36,147
Google Network	32,780	31,312	30,359
Google advertising	224,473	237,855	264,590
Google subscriptions, platforms, and devices	29,055	34,688	40,340
Google Services total	253,528	272,543	304,930
Google Cloud	26,280	33,088	43,229
Other Bets	1,068	1,527	1,648
Hedging gains (losses)	1,960	236	211
Total revenues	\$282,836	\$307,394	\$350,018

By Geography

Geography	2022	2023	2024
United States	\$134,814 (48%)	\$146,286 (47%)	\$170,447 (49%)
EMEA	82,062 (29%)	91,038 (30%)	102,127 (29%)
APAC	47,024 (16%)	51,514 (17%)	56,815 (16%)
Other Americas	16,976 (6%)	18,320 (6%)	20,418 (6%)
Hedging gains (losses)	1,960 (1%)	236 (0%)	211 (0%)
Total revenues	\$282,836 (100%)	\$307,394 (100%)	\$350,018 (100%)

2. Gains and Losses on Debt Securities (Note 3, Page 72)

This table details a component of "Other income (expense), net".

Description	2022	2023	2024
Unrealized gain (loss) on fair value option debt securities	\$(557)	\$386	\$30
Gross realized gain on debt securities	103	182	482
Gross realized loss on debt securities	(1,588)	(1,833)	(1,553)
(Increase) decrease in allowance for credit losses	(22)	50	(2)
Total gain (loss) on debt securities recognized in other income (expense), net	\$(2,064)	\$(1,215)	\$(1,043)

3. Gains and Losses on Equity Securities (Note 3, Page 73)

This table details a component of "Other income (expense), net".

Description	2022	2023	2024
Realized net gain (loss) on equity securities sold during the period	\$(442)	\$690	\$186
Unrealized net gain (loss) on marketable equity securities	(3,242)	790	156
Unrealized net gain (loss) on non-marketable equity securities	229	(1,088)	3,372
Total gain (loss) on equity securities in other income (expense), net	\$(3,455)	\$392	\$3,714

4. Derivative Gains/(Losses) in Income Statement (Note 3, Page 76)

This table details the impact of derivatives on "Revenues" and "Other income (expense), net".

Description	2022 Revenues	2022 OI&E	2023 Revenues	2023 OI&E	2024 Revenues	2024 OI&E
Effect of cash flow hedges:						
Foreign exchange contracts						
Amount reclassified from AOCI to income	\$2,046 (85)	\$0 0	\$213 24	\$0 0	\$174 37	\$0 0
Amount excluded from assessment of effectiveness (amortized)						
Effect of fair value hedges:						
Foreign exchange contracts						
Hedged items		(162)		59		(59)
Derivatives designated as hedging instruments	0	163 16	0	(59) 15	0	58 13
Amount excluded from assessment of effectiveness						
Effect of net investment hedges:						
Foreign exchange contracts						
Amount excluded from assessment of effectiveness	0	171	0	187	0	137
Effect of non designated hedges:						
Foreign exchange contracts		(395) 144		7 53		335 174
Other contracts	0		0		0	
Total gains (losses)	\$1,961	\$(63)	\$237	\$262	\$211	\$658

5. Components of Other Income (Expense), Net (Note 7, Page 83)

This table provides a full breakdown of the "Other income (expense), net" line item.

Component	2022	2023	2024
Interest income	\$2,174	\$3,865	\$4,482
Interest expense	(357)	(308)	(268)
Foreign currency exchange gain (loss), net	(654)	(1,238)	(409)
Gain (loss) on debt securities, net	(2,064)	(1,215)	(1,043)
Gain (loss) on equity securities, net	(3,455)	392	3,714
Performance fees	798	257	218
Income (loss) and impairment from equity method investments, net	(337)	(628)	(188)
Other	381	299	919
Other income (expense), net	\$(3,514)	\$1,424	\$7,425

6. Income Tax Reconciliation and Deferred Taxes (Note 14, Pages 85-86)

These tables provide details related to "Provision for income taxes".

Effective Tax Rate Reconciliation

Rate Component	2022	2023	2024
U.S. federal statutory tax rate	21.0%	21.0%	21.0%
Foreign income taxed at different rates	3.0	0.3	0.5
Foreign-derived intangible income deduction	(5.4)	(4.6)	(3.8)
Stock-based compensation expense	(1.2)	(0.8)	(1.5)
Federal research credit	(2.2)	(1.8)	(1.5)
Deferred tax asset valuation allowance	0.9	0.6	0.4
State and local income taxes	0.8	1.0	1.1
Effect of tax law change	0.0	(1.4)	0.0
Other	(1.0)	(0.4)	0.2
Effective tax rate	15.9%	13.9%	16.4%

Significant Components of Deferred Tax Assets and Liabilities (as of December 31)

Component	2023	2024
Deferred tax assets:		
Accrued employee benefits	\$1,855	\$1,834
Accruals and reserves not currently deductible	2,481	2,552
Tax credits	6,609	6,384
Net operating losses	2,965	3,472
Operating leases	3,526	3,336
Capitalized research and development	17,757	25,903
Other	1,951	1,376
Total deferred tax assets	37,144	44,857
Valuation allowance	(10,999)	(11,493)
Total deferred tax assets net of valuation allowance	26,145	33,364
Deferred tax liabilities:		
Property and equipment, net	(8,189)	(9,932)
Net investment gains	(2,405)	(2,978)
Operating leases	(2,965)	(2,986)
Other	(902)	(1,008)
Total deferred tax liabilities	(14,461)	(16,904)
Net deferred tax assets (liabilities)	\$11,684	\$16,460

7. Segment Revenue and Operating Income (Note 15, Page 96)

This table provides a breakdown of "Revenues" and "Income from operations" by reportable segment.

Segment	2022 Revenues	2023 Revenues	2024 Revenues	2022 Operating Income (Loss)	2023 Operating Income (Loss)	2024 Operating Income (Loss)
Google Services	\$253,528	\$272,543	\$304,930	\$82,699	\$95,858	\$121,263
Google Cloud	26,280	33,088	43,229	(1,922)	1,716	6,112
Other Bets	1,068	1,527	1,648	(4,636)	(4,095)	(4,444)
Alphabet-level activities	—	—	—	(1,299)	(9,186)	(10,541)
Hedging gains (losses)	1,960	236	211	—	—	—
Total	\$282,836	\$307,394	\$350,018	\$74,842	\$84,293	\$112,390

Step 5: Extracting relevant fields as JSON for ratio calculation

Then, from this final extracted information containing primary and supplementary tables, we extract the

relevant elements from each financial statement (Balance Sheet, Income Statement, or Cash Flows). When crafting the prompt, it is important to be careful so that it prevents LLM from hallucinating as much as possible. LLM is quite bad at doing math on the spot, but is much better at copying numbers. So, for retrieving fields that often requires summation of multiple elements, we ask LLM to copy the elements into an array instead of asking it to sum them up and return a single value. We use the following prompt:

```
STATEMENT_CONFIGS: Dict[str, Dict[str, Any]] = {
    "consolidated-balance-sheet": {
        "suffix": ".consolidated-balance-sheet.llm.json",
        "output_suffix": ".consolidated-balance-sheet.normalized.json",
        "fields": [
            "cash_and_cash_equivalents",
            "marketable_securities",
            "accounts_receivable",
            "total_current_liabilities",
            "total_debt_short_term_and_long_term",
            "total_equity",
            "total_assets",
        ],
    },
    "consolidated-income-statement": {
        "suffix": ".consolidated-income-statement.llm.json",
        "output_suffix": ".consolidated-income-statement.normalized.json",
        "fields": [
            "revenue",
            "total_operating_cost",
            "net_income",
            "taxes",
            "interest_expenses",
        ],
    },
    "consolidated-cash-flow-statement": {
        "suffix": ".consolidated-cash-flow-statement.llm.json",
        "output_suffix": ".consolidated-cash-flow-statement.normalized.json",
        "fields": [
            "depreciation_and_amortization",
        ],
    },
}

fields_schema = ",\n".join(
    [f' "{field}": number[]'
     for field in required_fields]
)

prompt = (
    "You are a financial statement extraction engine.\n"
    f"Extract normalized {statement_type} values from the "
    f"statement and supplementary tables.\n\n"
    "Return ONLY valid JSON with this exact schema:\n"
    "{\n"
    '  "company_id": "string",\n'
    '  "periods": [\n'
    '    {\n'
    '      "year": "string",\n'
    '      "currency": "string",\n'
    '      "values": {\n'
    f"        {fields_schema}\n"
    '      }\n'
    "    }\n"
    "  ]\n"
    "}"
)
```

```

"    }\n"
" ],\n"
' "notes": string\n'
"}\n\n"
"Rules:\n"
"1) Use every period/column available in the statement.\n"
"2) Return numeric array values only (no commas, no currency "
"symbols, no percent signs).\n"
"3) If a value cannot be directly mapped to a field, try to "
"map number(s) to the corresponding field by taking into "
"account the industry the company is in, and note what you "
"did in notes field. If you cannot find a match, return an "
"empty array.\n"
"4) If multiple numbers make up a field, return them in an "
"array without adding them up.\n"
"5) If there are multiple close synonyms, use best "
"accounting match.\n"
"6) For parentheses negatives, return negative numbers.\n"
"7) For year, report the year of the period only.\n"
"8) For currency, report its formal 3-letter acronym.\n"
"9) For total_operating_cost, list all elements that should "
"be included in standard practice for the industry the "
"company is in. Elements that should be subtracted should "
"be negative. The sign convention should be such that if an "
"element is a cost, it should be negative, and if it is an "
"income, it should be positive.\n"
"10) Return JSON only.\n\n"
f"company_id: {company_id}\n\n"
"statement and supplementary tables:\n"
f"{statement_and_supplementary_tables}\n"
)

```

Where

- `required_fields` are selected from `STATEMENT_CONFIGS` and are dependent on the type of financial statement we are feeding the LLM
- `statement_type` is one of ["Consolidated Balance Sheet", "Consolidated Income Statement", "Consolidated Cash Flow Statement"]
- `statement_text` is the extracted primary and supplementary tables from Step 4

The LLM struggled to identify the correct elements for marketable securities. So, we add the definition of marketable securities according to definition:

- `short_term_market_securities`: these are liquid, unrestricted debt or equity investments intended to be sold in the short term, e.g., U.S. Treasuries, commercial paper, money market funds, publicly traded equities, short-term investments.

The LLM kept wanting to add up the components before reporting them in a single field, e.g., it kept calculating the total debt by adding up the short-term and long-term borrowings often with hallucination and incorrect summed results.

So, we had to explicitly tell LLM to return the original source label and its value as an array of maps:

```

fields_schema = ",\n".join(
    [
        f'  "{field}": [{{"source_label": number}}]'
        for field in required_fields
    ]
)

```

```

)

prompt = (
    "You are a financial statement extraction engine.\n"
    f"Extract normalized {statement_type} values from the "
    f"statement and supplementary tables.\n\n"
    "Return ONLY valid JSON with this exact schema:\n"
    "{\n"
    '  "company_id": "string",\n'
    '  "periods": [\n'
    "    {\n"
    '      "year": "string",\n'
    '      "currency": "string",\n'
    '      "scale": number,\n'
    '      "values": {\n'
    f"        {fields_schema}\n"
    "      }\n"
    "    },\n"
    '  ],\n'
    '  "notes": string\n'
    "}\n\n"
    "Rules:\n"
    "1) Use every period/column available in the statement.\n"
    "2) Return an array of maps that make up each field where "
    "each map is {source_label: number} (source_label = actual "
    "label from the statement; no commas, no currency symbols, "
    "no percent signs in numbers).\n"
    "3) Ensure that the elements in the array correctly make up "
    "the total value of the field.\n"
    "4) If multiple elements need to be combined to make up a "
    "field, return all of them individually in an array.\n"
    "5) If a value(s) cannot be directly mapped to a field, "
    "try to map element(s) to the corresponding field by "
    "taking into account the industry the company is in, and "
    "note your reasoning in the notes field. If you cannot "
    "find a match, return an empty array.\n"
    "6) If there are multiple close synonyms, use best "
    "accounting match.\n"
    "7) For year, report the year of the period only.\n"
    "8) For currency, report its formal 3-letter acronym.\n"
    "9) For scale, report the scale of the values in the "
    "statement. For example, if the values are in millions, "
    "the scale should be 1E6.\n"
    "10) For total_operating_cost, list all elements that "
    "reflect the total operational costs incurred in "
    "generating income and are included in standard practice "
    "for the industry the company is in.\n"
    "11) For taxes, it is the tax assessed on the income. If "
    "a tax belongs to the cost of revenue, it should be part "
    "of total_operating_cost.\n"
    "12) Generally, numbers in parentheses are negative.\n"
    "13) For taxes, interest_expenses, and "
    "total_operating_cost, the sign convention is the "
    "opposite: if an element reduces income, it should be "
    "positive; and if it increases income, it should be "
    "negative.\n"
    "14) For short_term_market_securities, these are liquid, "
    "unrestricted debt or equity investments intended to be "
    "sold in the short term, e.g., U.S. Treasuries, "

```

```

"commercial paper, money market funds, publicly traded "
"equities, short-term investments.\n"
"15) Return JSON only.\n\n"
f"company_id: {company_id}\n\n"
"statement and supplementary tables:\n"
f"{statement_and_supplementary_tables}\n"
)

```

Output example for GM's 2023 annual report:

Before: LLM attempts to add the total debt by itself

```

{
  "year": "2023",
  "currency": "USD",
  "values": {
    "cash_and_cash_equivalents": [18853.0],
    "short_term_market_securities": [7613.0],
    "accounts_receivable": [12378.0],
    "total_current_liabilities": [94445.0],
    "total_debt_short_term_and_long_term": [16413.0, 105327.0],
    "total_equity": [68189.0],
    "total_assets": [273064.0]
  }
},

```

After: LLM lists out all the original elements without summation

```

{
  "year": "2023",
  "currency": "USD",
  "scale": 1000000,
  "values": {
    "cash_and_cash_equivalents": [
      {"Cash and cash equivalents": 18853.0}
    ],
    "short_term_market_securities": [
      {"Marketable debt securities": 7613.0}
    ],
    "accounts_receivable": [
      {"Accounts and notes receivable, net": 12378.0},
      {"GM Financial receivables, net (current)": 39076.0}
    ],
    "total_current_liabilities": [
      {"Total current liabilities": 94445.0}
    ],
    "total_debt_short_term_and_long_term": [
      {"Short-term debt and current portion of long-term debt - Automotive": 428.0},
      {"Short-term debt and current portion of long-term debt - GM Financial": 38540.0},
      {"Long-term debt - Automotive": 15985.0},
      {"Long-term debt - GM Financial": 66788.0}
    ],
    "total_equity": [
      {"Total Equity": 68189.0}
    ],
    "total_assets": [
      {"Total Assets": 273064.0}
    ]
  }
}

```

```
}  
},  
...
```

Even with clever prompt engineering, LLM still hallucinates. To measure its hallucination rate, we repeated Step 5 (i.e., extracting json from financial statements for financial ratio calculations) for 101 runs and plotted each field value vs run using the following scripts:

Run `llm_extract_json_from_financial_statements.py` with these arguments:

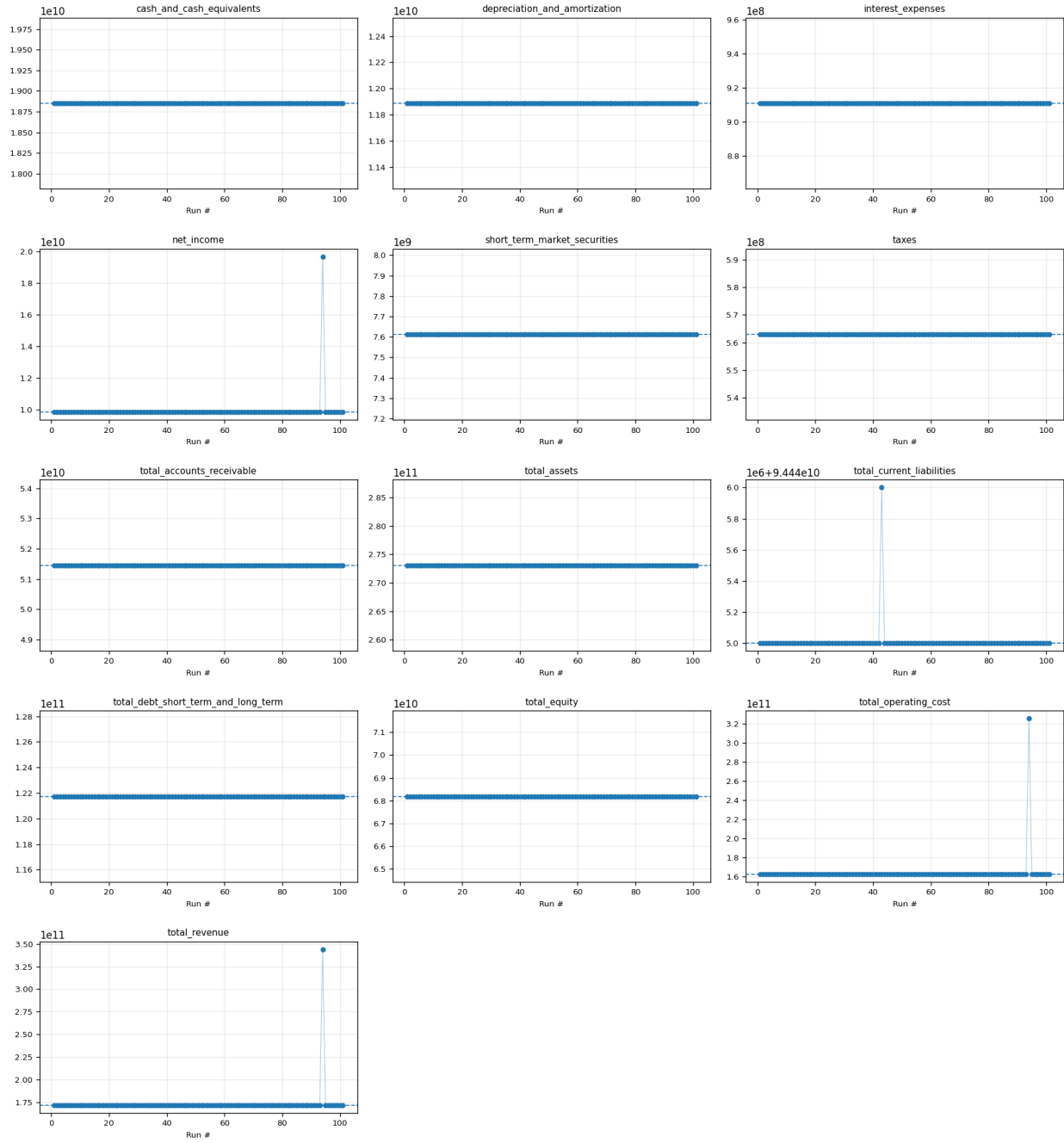
```
python llm_extract_json_from_financial_statements.py \  
--input-dir extracted_text \  
--num-extraction-runs 101 \  
--runs-output-dir deepseek_financial_statements_runs \  
--max-workers 9 \  
--plot-distributions
```

Then, run a simple script, `calculate_hallucination_rates.py`, to plot the extracted json data to visualize hallucination rates for different financial fields from different companies:

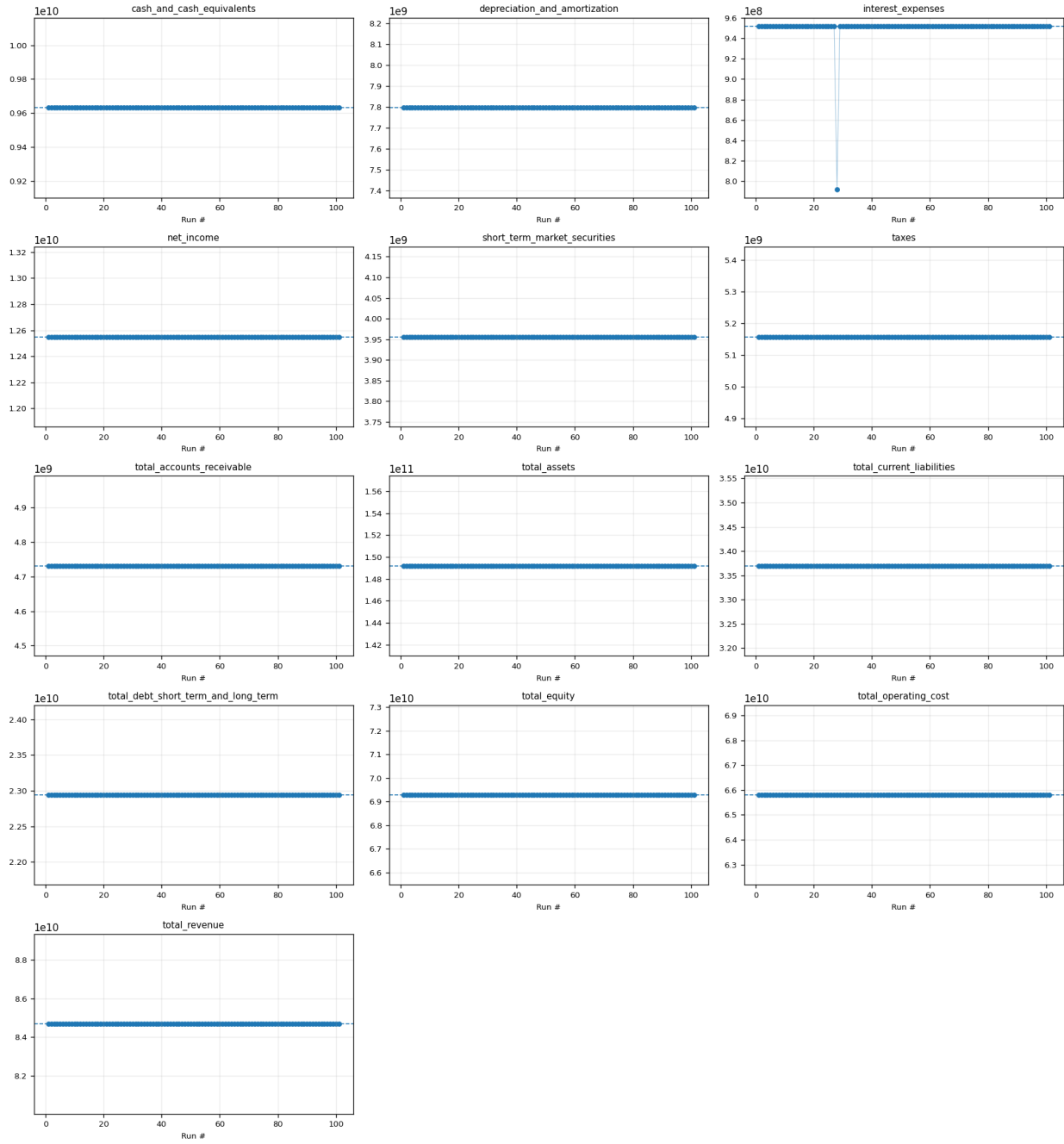
```
python calculate_hallucination_rates.py \  
--run-values-file deepseek_financial_statements_runs/  
runs_3_with_operating_cost_for_generating_income/  
extraction_run_values.json \  
--output-file hallucination_rates.json \  
--top-k 10
```

The results are reproduced below for GM, LVMH, Tencent, Alibaba, JPMorgan Chase, Exxon Mobil, Volkswagen, Microsoft, and Google:

2023 General Motors Annual Report - 2023 field distributions



lvmh_dec_2024 - 2024 field distributions



tencent_2024 - 2024 field distributions



alibaba_2025 - 2025 field distributions



jpmorgan_2024 - 2024 field distributions



exxon_2024 - 2024 field distributions



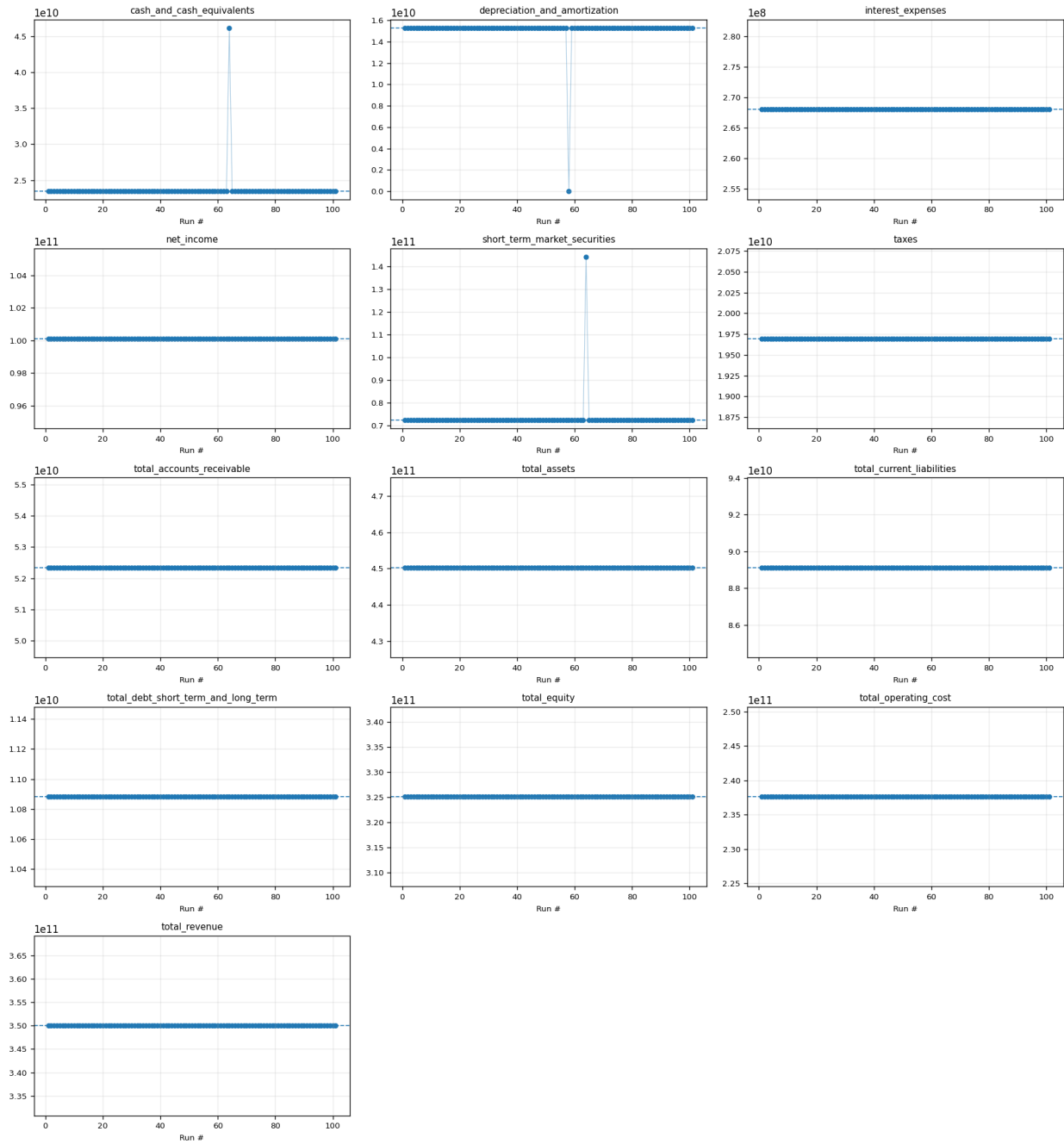
volkswagen_2024 - 2024 field distributions



microsoft_2025 - 2025 field distributions



google_2024 - 2024 field distributions



We also calculate the top 15 hallucinated fields as below:

```
- exxon_2024 | total_operating_cost | rate_non_null=0.2970 (90/303)
- microsoft_2025 | depreciation_and_amortization | rate_non_null=0.1980 (60/303)
- volkswagen_2024 | total_accounts_receivable | rate_non_null=0.1881 (38/202)
- jpmorgan_2024 | depreciation_and_amortization | rate_non_null=0.1683 (51/303)
- volkswagen_2024 | depreciation_and_amortization | rate_non_null=0.1683 (34/202)
- volkswagen_2024 | total_operating_cost | rate_non_null=0.1287 (26/202)
```

```

- alibaba_2025 | cash_and_cash_equivalents | rate_non_null=0.0990 (20/202)
- alibaba_2025 | short_term_market_securities | rate_non_null=0.0990 (20/202)
- alibaba_2025 | total_accounts_receivable | rate_non_null=0.0990 (20/202)
- alibaba_2025 | total_assets | rate_non_null=0.0990 (20/202)
- alibaba_2025 | total_current_liabilities | rate_non_null=0.0990 (20/202)
- alibaba_2025 | total_debt_short_term_and_long_term | rate_non_null=0.0990 (20/202)
- alibaba_2025 | total_equity | rate_non_null=0.0990 (20/202)
- tencent_2024 | short_term_market_securities | rate_non_null=0.0990 (20/202)
- jpmorgan_2024 | total_operating_cost | rate_non_null=0.0792 (24/303)

```

As you can see, the hallucination is much better than a coin-toss; LLM retrieves the correct values more than 70% of the time. Thus, it is important to repeat the extraction step multiple times to retrieve the correct values to do any further quantitative analysis.

The most hallucinated value was ExxonMobile’s operating cost because LLM was not sure whether to include “Other taxes and duties” as part of operating cost. It reasoned more often than not that it is part of the operating cost. There are several ways to reduce this hallucination rate. One is to give direct examples of what we want for this specific item for this company. Another is to reword the name or the definition of “Operating Cost” in a way that there are less ambiguities for LLM in determining the elements that belong to this field.

As an example of this second point, we were able to reduce the hallucination rate in extracting the Short-Term Market Securities from JPMorgan Chase’s annual report. Previously, we used the following name and definition in the LLM prompt:

For marketable_securities, these are most liquid, unrestricted debt or equity investments intended to be sold in the near term (e.g., U.S. Treasuries, commercial paper, money market funds, publicly traded equities)

And we observed high hallucination rate for this field because LLM outputted different array of elements that made up marketable securities, as evidenced by the following 3 extraction runs:

Run 1:

```

"marketable_securities": [
  {"Federal funds sold and securities purchased under resale agreements": 295001.0},
  {"Securities borrowed": 219546.0},
  {"Available-for-sale securities": 406852.0}
],

```

Run 2:

```

"marketable_securities": [
  {"Federal funds sold and securities purchased under resale agreements": 295001.0},
  {"Securities borrowed": 219546.0},
  {"Trading assets": 637784.0},
  {"Available-for-sale securities": 406852.0}
],

```

Run 5:

```

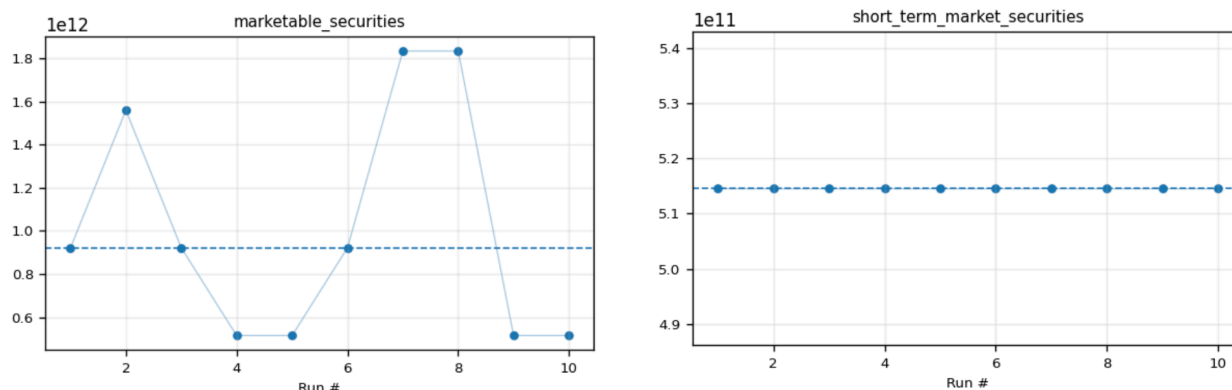
"marketable_securities": [
  {"Federal funds sold and securities purchased under resale agreements": 295001.0},
  {"Securities borrowed": 219546.0}
],

```

Then we updated our naming and definition from “marketable_securities” to “short_term_market_securities”:
For short_term_market_securities, these are liquid, unrestricted debt or equity investments intended to

be sold in the short term, e.g., U.S. Treasuries, commercial paper, money market funds, publicly traded equities, short-term investments.

We see that the hallucination rate practically goes to zero:



After obtaining the correct field values, calculating the various financial ratios is straightforward. We write a Python script for the calculation, which gives us definite results. The script is part of this json extraction script that runs automatically after extracting the relevant fields: `llm_extract_json_from_financial_statements.py`

Calculated fields and financial ratios for GM are as follows for 2022 and 2023 in the 2023 annual report:

```
{
  "year": "2022",
  "currency": "USD",
  "field_values": {
    "cash_and_cash_equivalents": 19153000000.0,
    "short_term_market_securities": 12150000000.0,
    "total_accounts_receivable": 46956000000.0,
    "total_current_liabilities": 91173000000.0,
    "total_debt_short_term_and_long_term": 114699000000.0,
    "total_equity": 71927000000.0,
    "total_assets": 264037000000.0,
    "total_revenue": 156735000000.0,
    "total_operating_cost": 146421000000.0,
    "net_income": 9708000000.0,
    "taxes": 1888000000.0,
    "interest_expenses": 987000000.0,
    "depreciation_and_amortization": 11290000000.0
  },
  "derived_values": {
    "ebit": 12583000000.0,
    "ebitda": 23873000000.0,
    "quick_assets": 78259000000.0,
    "debt_plus_equity": 186626000000.0
  },
  "ratios": {
    "cost_to_income_ratio": 0.9341946597760551,
    "quick_ratio": 0.8583571890801005,
    "debt_to_equity_ratio": 1.5946584731741904,
    "debt_to_assets_ratio": 0.43440502656824614,
    "debt_to_capital_ratio": 0.6145928220076516,
    "debt_to_ebitda_ratio": 4.804549072173585,
    "interest_coverage_ratio": 12.74873353596758
  }
},
{
```



```

"year": "2023",
"currency": "USD",
"field_values": {
  "cash_and_cash_equivalents": 18853000000.0,
  "short_term_market_securities": 7613000000.0,
  "total_accounts_receivable": 51454000000.0,
  "total_current_liabilities": 94445000000.0,
  "total_debt_short_term_and_long_term": 121741000000.0,
  "total_equity": 68189000000.0,
  "total_assets": 273064000000.0,
  "total_revenue": 171842000000.0,
  "total_operating_cost": 162544000000.0,
  "net_income": 9840000000.0,
  "taxes": 563000000.0,
  "interest_expenses": 911000000.0,
  "depreciation_and_amortization": 11888000000.0
},
"derived_values": {
  "ebit": 11314000000.0,
  "ebitda": 23202000000.0,
  "quick_assets": 77920000000.0,
  "debt_plus_equity": 189930000000.0
},
"ratios": {
  "cost_to_income_ratio": 0.9458921567486412,
  "quick_ratio": 0.8250304409974059,
  "debt_to_equity_ratio": 1.7853466101570634,
  "debt_to_assets_ratio": 0.44583321126182873,
  "debt_to_capital_ratio": 0.640978255146633,
  "debt_to_ebitda_ratio": 5.247004568571675,
  "interest_coverage_ratio": 12.419319429198683
}
}

```

h, i) Applicability to Other Annual Reports

Our script works for other annual reports as well, including LVMH, Tencent, Alibaba, JPMorgan Chase, ExxonMobil, Volkswagen, Microsoft, and Google:

LVMH:

```

{
  "year": "2022",
  "currency": "EUR",
  "field_values": {
    "cash_and_cash_equivalents": 7301000000.0,
    "short_term_market_securities": 3552000000.0,
    "total_accounts_receivable": 4258000000.0,
    "total_current_liabilities": 31543000000.0,
    "total_debt_short_term_and_long_term": 19739000000.0,
    "total_equity": 56604000000.0,
    "total_assets": 134646000000.0,
    "total_revenue": 79184000000.0,
    "total_operating_cost": 58220000000.0,
    "net_income": 14084000000.0,
    "taxes": 5362000000.0,
    "interest_expenses": 271000000.0,
    "depreciation_and_amortization": 6226000000.0
  },

```

```

"derived_values": {
  "ebit": 19717000000.0,
  "ebitda": 25943000000.0,
  "quick_assets": 15111000000.0,
  "debt_plus_equity": 76343000000.0
},
"ratios": {
  "cost_to_income_ratio": 0.7352495453626995,
  "quick_ratio": 0.4790603303427068,
  "debt_to_equity_ratio": 0.3487209384495795,
  "debt_to_assets_ratio": 0.14659923057498922,
  "debt_to_capital_ratio": 0.2585567766527383,
  "debt_to_ebitda_ratio": 0.7608603476853101,
  "interest_coverage_ratio": 72.75645756457564
}
},
{
  "year": "2023",
  "currency": "EUR",
  "field_values": {
    "cash_and_cash_equivalents": 7774000000.0,
    "short_term_market_securities": 3490000000.0,
    "total_accounts_receivable": 4728000000.0,
    "total_current_liabilities": 33145000000.0,
    "total_debt_short_term_and_long_term": 21907000000.0,
    "total_equity": 62701000000.0,
    "total_assets": 143694000000.0,
    "total_revenue": 86153000000.0,
    "total_operating_cost": 63600000000.0,
    "net_income": 15174000000.0,
    "taxes": 5673000000.0,
    "interest_expenses": 760000000.0,
    "depreciation_and_amortization": 7177000000.0
  },
  "derived_values": {
    "ebit": 21607000000.0,
    "ebitda": 28784000000.0,
    "quick_assets": 15992000000.0,
    "debt_plus_equity": 84608000000.0
  },
  "ratios": {
    "cost_to_income_ratio": 0.7382215361043725,
    "quick_ratio": 0.4824860461608086,
    "debt_to_equity_ratio": 0.3493883670116904,
    "debt_to_assets_ratio": 0.1524559132601222,
    "debt_to_capital_ratio": 0.2589235060514372,
    "debt_to_ebitda_ratio": 0.7610825458588104,
    "interest_coverage_ratio": 28.430263157894736
  }
}
},
{
  "year": "2024",
  "currency": "EUR",
  "field_values": {
    "cash_and_cash_equivalents": 9631000000.0,
    "short_term_market_securities": 3956000000.0,
    "total_accounts_receivable": 4731000000.0,
    "total_current_liabilities": 33696000000.0,
    "total_debt_short_term_and_long_term": 22942000000.0,

```

```

    "total_equity": 69287000000.0,
    "total_assets": 149190000000.0,
    "total_revenue": 84683000000.0,
    "total_operating_cost": 65804000000.0,
    "net_income": 12550000000.0,
    "taxes": 5157000000.0,
    "interest_expenses": 952000000.0,
    "depreciation_and_amortization": 7796000000.0
  },
  "derived_values": {
    "ebit": 18659000000.0,
    "ebitda": 26455000000.0,
    "quick_assets": 18318000000.0,
    "debt_plus_equity": 92229000000.0
  },
  "ratios": {
    "cost_to_income_ratio": 0.7770626926301619,
    "quick_ratio": 0.5436253561253561,
    "debt_to_equity_ratio": 0.3311155050731017,
    "debt_to_assets_ratio": 0.15377706280581807,
    "debt_to_capital_ratio": 0.24875039304340282,
    "debt_to_ebitda_ratio": 0.8672084672084672,
    "interest_coverage_ratio": 19.599789915966387
  }
}

```

Tencent:

```

{
  "year": "2023",
  "currency": "CNY",
  "field_values": {
    "cash_and_cash_equivalents": 172320000000.0,
    "short_term_market_securities": 14903000000.0,
    "total_accounts_receivable": 46606000000.0,
    "total_current_liabilities": 352157000000.0,
    "total_debt_short_term_and_long_term": 348618000000.0,
    "total_equity": 873681000000.0,
    "total_assets": 1577246000000.0,
    "total_revenue": 609015000000.0,
    "total_operating_cost": 453642000000.0,
    "net_income": 118048000000.0,
    "taxes": 43276000000.0,
    "interest_expenses": 12268000000.0,
    "depreciation_and_amortization": 59008000000.0
  },
  "derived_values": {
    "ebit": 173592000000.0,
    "ebitda": 232600000000.0,
    "quick_assets": 233829000000.0,
    "debt_plus_equity": 1222299000000.0
  },
  "ratios": {
    "cost_to_income_ratio": 0.744878204970321,
    "quick_ratio": 0.6639907768410113,
    "debt_to_equity_ratio": 0.3990220686955536,
    "debt_to_assets_ratio": 0.22102956672579926,
    "debt_to_capital_ratio": 0.285214992403659,

```

```

    "debt_to_ebitda_ratio": 1.4987876182287188,
    "interest_coverage_ratio": 14.149983697424194
  }
},
{
  "year": "2024",
  "currency": "CNY",
  "field_values": {
    "cash_and_cash_equivalents": 132519000000.0,
    "short_term_market_securities": 12913000000.0,
    "total_accounts_receivable": 48203000000.0,
    "total_current_liabilities": 396909000000.0,
    "total_debt_short_term_and_long_term": 338615000000.0,
    "total_equity": 1053896000000.0,
    "total_assets": 1780995000000.0,
    "total_revenue": 660257000000.0,
    "total_operating_cost": 460160000000.0,
    "net_income": 196467000000.0,
    "taxes": 45018000000.0,
    "interest_expenses": 11981000000.0,
    "depreciation_and_amortization": 56213000000.0
  },
  "derived_values": {
    "ebit": 253466000000.0,
    "ebitda": 309679000000.0,
    "quick_assets": 193635000000.0,
    "debt_plus_equity": 1392511000000.0
  },
  "ratios": {
    "cost_to_income_ratio": 0.6969407367131284,
    "quick_ratio": 0.4878574181991338,
    "debt_to_equity_ratio": 0.3212983064742631,
    "debt_to_assets_ratio": 0.19012686728486042,
    "debt_to_capital_ratio": 0.24316863565171118,
    "debt_to_ebitda_ratio": 1.093438689740021,
    "interest_coverage_ratio": 21.155663133294382
  }
}
}

```

Alibaba:

```

{
  "year": "2024",
  "currency": "CNY",
  "field_values": {
    "cash_and_cash_equivalents": 248125000000.0,
    "short_term_market_securities": 262955000000.0,
    "total_accounts_receivable": 29362000000.0,
    "total_current_liabilities": 421507000000.0,
    "total_debt_short_term_and_long_term": 170776000000.0,
    "total_equity": 1101871000000.0,
    "total_assets": 1764829000000.0,
    "total_revenue": 941168000000.0,
    "total_operating_cost": 827818000000.0,
    "net_income": 71332000000.0,
    "taxes": 22529000000.0,
    "interest_expenses": 7947000000.0,
    "depreciation_and_amortization": 44504000000.0
  }
}

```

```

    },
    "derived_values": {
      "ebit": 101808000000.0,
      "ebitda": 146312000000.0,
      "quick_assets": 540442000000.0,
      "debt_plus_equity": 1272647000000.0
    },
    "ratios": {
      "cost_to_income_ratio": 0.8795645410808697,
      "quick_ratio": 1.282166132472296,
      "debt_to_equity_ratio": 0.1549872898007117,
      "debt_to_assets_ratio": 0.0967663156033814,
      "debt_to_capital_ratio": 0.13418960638731714,
      "debt_to_ebitda_ratio": 1.1672043304718684,
      "interest_coverage_ratio": 12.810872027180068
    }
  },
  {
    "year": "2025",
    "currency": "CNY",
    "field_values": {
      "cash_and_cash_equivalents": 145487000000.0,
      "short_term_market_securities": 228826000000.0,
      "total_accounts_receivable": 30652000000.0,
      "total_current_liabilities": 435346000000.0,
      "total_debt_short_term_and_long_term": 230703000000.0,
      "total_equity": 1078393000000.0,
      "total_assets": 1804227000000.0,
      "total_revenue": 996347000000.0,
      "total_operating_cost": 856203000000.0,
      "net_income": 125976000000.0,
      "taxes": 35445000000.0,
      "interest_expenses": 9596000000.0,
      "depreciation_and_amortization": 42459000000.0
    },
    "derived_values": {
      "ebit": 171017000000.0,
      "ebitda": 213476000000.0,
      "quick_assets": 404965000000.0,
      "debt_plus_equity": 1309096000000.0
    },
    "ratios": {
      "cost_to_income_ratio": 0.8593421769724805,
      "quick_ratio": 0.9302141285322479,
      "debt_to_equity_ratio": 0.21393221209707408,
      "debt_to_assets_ratio": 0.12786805651395305,
      "debt_to_capital_ratio": 0.17623077299143836,
      "debt_to_ebitda_ratio": 1.0806975959826866,
      "interest_coverage_ratio": 17.821696540225094
    }
  }
}

```

JPMorgan Chase:

```

{
  "year": "2023",
  "currency": "USD",
  "field_values": {

```

```

    "cash_and_cash_equivalents": 624151000000.0,
    "short_term_market_securities": 476588000000.0,
    "total_accounts_receivable": 107363000000.0,
    "total_current_liabilities": 3155690000000.0,
    "total_debt_short_term_and_long_term": 436537000000.0,
    "total_equity": 327878000000.0,
    "total_assets": 3875393000000.0,
    "total_revenue": 239425000000.0,
    "total_operating_cost": 177813000000.0,
    "net_income": 49552000000.0,
    "taxes": 12060000000.0,
    "interest_expenses": 81321000000.0,
    "depreciation_and_amortization": 7512000000.0
  },
  "derived_values": {
    "ebit": 142933000000.0,
    "ebitda": 150445000000.0,
    "quick_assets": 1208102000000.0,
    "debt_plus_equity": 764415000000.0
  },
  "ratios": {
    "cost_to_income_ratio": 0.7426668058891093,
    "quick_ratio": 0.38283291451314927,
    "debt_to_equity_ratio": 1.3314007039203606,
    "debt_to_assets_ratio": 0.11264328546808026,
    "debt_to_capital_ratio": 0.5710733044223361,
    "debt_to_ebitda_ratio": 2.90163847253149,
    "interest_coverage_ratio": 1.7576394781175835
  }
},
{
  "year": "2024",
  "currency": "USD",
  "field_values": {
    "cash_and_cash_equivalents": 469317000000.0,
    "short_term_market_securities": 514547000000.0,
    "total_accounts_receivable": 101223000000.0,
    "total_current_liabilities": 3256638000000.0,
    "total_debt_short_term_and_long_term": 454311000000.0,
    "total_equity": 344758000000.0,
    "total_assets": 4002814000000.0,
    "total_revenue": 278906000000.0,
    "total_operating_cost": 203825000000.0,
    "net_income": 58471000000.0,
    "taxes": 16610000000.0,
    "interest_expenses": 101350000000.0,
    "depreciation_and_amortization": 7938000000.0
  },
  "derived_values": {
    "ebit": 176431000000.0,
    "ebitda": 184369000000.0,
    "quick_assets": 1085087000000.0,
    "debt_plus_equity": 799069000000.0
  },
  "ratios": {
    "cost_to_income_ratio": 0.7308017755085943,
    "quick_ratio": 0.33319239043455245,
    "debt_to_equity_ratio": 1.3177678255471954,
    "debt_to_assets_ratio": 0.11349790422437815,

```

```

    "debt_to_capital_ratio": 0.5685504005286152,
    "debt_to_ebitda_ratio": 2.464139849974779,
    "interest_coverage_ratio": 1.740809077454366
  }
}

```

ExxonMobil:

```

{
  "year": "2023",
  "currency": "USD",
  "field_values": {
    "cash_and_cash_equivalents": 31568000000.0,
    "short_term_market_securities": 0.0,
    "total_accounts_receivable": 38015000000.0,
    "total_current_liabilities": 65316000000.0,
    "total_debt_short_term_and_long_term": 41573000000.0,
    "total_equity": 212538000000.0,
    "total_assets": 376317000000.0,
    "total_revenue": 344582000000.0,
    "total_operating_cost": 290950000000.0,
    "net_income": 36010000000.0,
    "taxes": 15429000000.0,
    "interest_expenses": 849000000.0,
    "depreciation_and_amortization": 20641000000.0
  },
  "derived_values": {
    "ebit": 52288000000.0,
    "ebitda": 72929000000.0,
    "quick_assets": 69583000000.0,
    "debt_plus_equity": 254111000000.0
  },
  "ratios": {
    "cost_to_income_ratio": 0.8443563505928923,
    "quick_ratio": 1.065328556555821,
    "debt_to_equity_ratio": 0.19560266869924436,
    "debt_to_assets_ratio": 0.11047335092488514,
    "debt_to_capital_ratio": 0.163601733100889,
    "debt_to_ebitda_ratio": 0.5700475805235229,
    "interest_coverage_ratio": 61.58775029446407
  }
},
{
  "year": "2024",
  "currency": "USD",
  "field_values": {
    "cash_and_cash_equivalents": 23187000000.0,
    "short_term_market_securities": 0.0,
    "total_accounts_receivable": 43681000000.0,
    "total_current_liabilities": 70307000000.0,
    "total_debt_short_term_and_long_term": 41710000000.0,
    "total_equity": 270606000000.0,
    "total_assets": 453475000000.0,
    "total_revenue": 349585000000.0,
    "total_operating_cost": 299716000000.0,
    "net_income": 33680000000.0,
    "taxes": 13810000000.0,
    "interest_expenses": 996000000.0,

```

```

    "depreciation_and_amortization": 23442000000.0
  },
  "derived_values": {
    "ebit": 48486000000.0,
    "ebitda": 71928000000.0,
    "quick_assets": 66868000000.0,
    "debt_plus_equity": 312316000000.0
  },
  "ratios": {
    "cost_to_income_ratio": 0.8573479983408899,
    "quick_ratio": 0.951085951612215,
    "debt_to_equity_ratio": 0.1541355328411048,
    "debt_to_assets_ratio": 0.09197860962566845,
    "debt_to_capital_ratio": 0.1335506346136605,
    "debt_to_ebitda_ratio": 0.5798854409965521,
    "interest_coverage_ratio": 48.68072289156626
  }
}

```

Volkswagen:

```

{
  "year": "2023",
  "currency": "EUR",
  "field_values": {
    "cash_and_cash_equivalents": 43449000000.0,
    "short_term_market_securities": 26772000000.0,
    "total_accounts_receivable": 88230000000.0,
    "total_current_liabilities": 206036000000.0,
    "total_debt_short_term_and_long_term": 232799000000.0,
    "total_equity": 189186000000.0,
    "total_assets": 600649000000.0,
    "total_revenue": 322284000000.0,
    "total_operating_cost": 314907000000.0,
    "net_income": 17861000000.0,
    "taxes": 5237000000.0,
    "interest_expenses": 3640000000.0,
    "depreciation_and_amortization": 27566000000.0
  },
  "derived_values": {
    "ebit": 26738000000.0,
    "ebitda": 54304000000.0,
    "quick_assets": 158451000000.0,
    "debt_plus_equity": 421985000000.0
  },
  "ratios": {
    "cost_to_income_ratio": 0.9771102505864393,
    "quick_ratio": 0.7690452153992506,
    "debt_to_equity_ratio": 1.2305297432156714,
    "debt_to_assets_ratio": 0.3875791019380703,
    "debt_to_capital_ratio": 0.5516760074410227,
    "debt_to_ebitda_ratio": 4.286958603417796,
    "interest_coverage_ratio": 7.345604395604395
  }
},
{
  "year": "2024",
  "currency": "EUR",

```



```

"field_values": {
  "cash_and_cash_equivalents": 40296000000.0,
  "short_term_market_securities": 27326000000.0,
  "total_accounts_receivable": 89985000000.0,
  "total_current_liabilities": 217039000000.0,
  "total_debt_short_term_and_long_term": 254081000000.0,
  "total_equity": 196731000000.0,
  "total_assets": 632905000000.0,
  "total_revenue": 324656000000.0,
  "total_operating_cost": 320570000000.0,
  "net_income": 12394000000.0,
  "taxes": 4411000000.0,
  "interest_expenses": 3446000000.0,
  "depreciation_and_amortization": 31346000000.0
},
"derived_values": {
  "ebit": 20251000000.0,
  "ebitda": 51597000000.0,
  "quick_assets": 157607000000.0,
  "debt_plus_equity": 450812000000.0
},
"ratios": {
  "cost_to_income_ratio": 0.9874143709033562,
  "quick_ratio": 0.7261690295292551,
  "debt_to_equity_ratio": 1.2915148095622957,
  "debt_to_assets_ratio": 0.40145203466555013,
  "debt_to_capital_ratio": 0.5636074461194467,
  "debt_to_ebitda_ratio": 4.924336686241448,
  "interest_coverage_ratio": 5.876668601276843
}
}

```

Microsoft:

```

{
  "year": "2024",
  "currency": "USD",
  "field_values": {
    "cash_and_cash_equivalents": 18315000000.0,
    "short_term_market_securities": 57228000000.0,
    "total_accounts_receivable": 56924000000.0,
    "total_current_liabilities": 125286000000.0,
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Google:

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